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23.	SIM/TF card
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25.	Touch Interface
26.	Test Point/GND/Shields
27.	WTR1605L TX
28.	WTR1605L RX
29.	WTR1605L POWER
30.	WTR1605L POWER DISTRIBUTION
31.	LTE/W/TD/GSM Antenna Switch
32.	TRX_Front-High Band QFE2340
33.	TRX_Front-Low Band QFE2320
34.	RESERVED
35.	TRX_UMTS_B1/2/5/8
36.	PRX_B34/39/3
37.	PRX_LTE_HB
38.	DRX_Antenna Switch
39.	QFE1550 DRX Tunner
40.	B7/40/41 DRX Switch
41.	B39/B1/B3 DRX Switch
42.	WTR2605_SECONDARY PATH
43.	WTR2605_POWER DISTRIBUTION
44.	Antenna_SECONDARY PATH
45.	RESERVED
46.	RESERVED
47.	CDMA BC0(Voice) TRX
48.	ET_APT
49.	WCN3620
50.	WIFI FEM
51.	GPS/X0 DISTRIBUTOR
52.	NFC

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Revision History

SCHEMATIC MCN	HW VERSION	VERSION	DESCRIPTION OF CHANGE
	EVT1.0	A	EVT1.0 Initial Release

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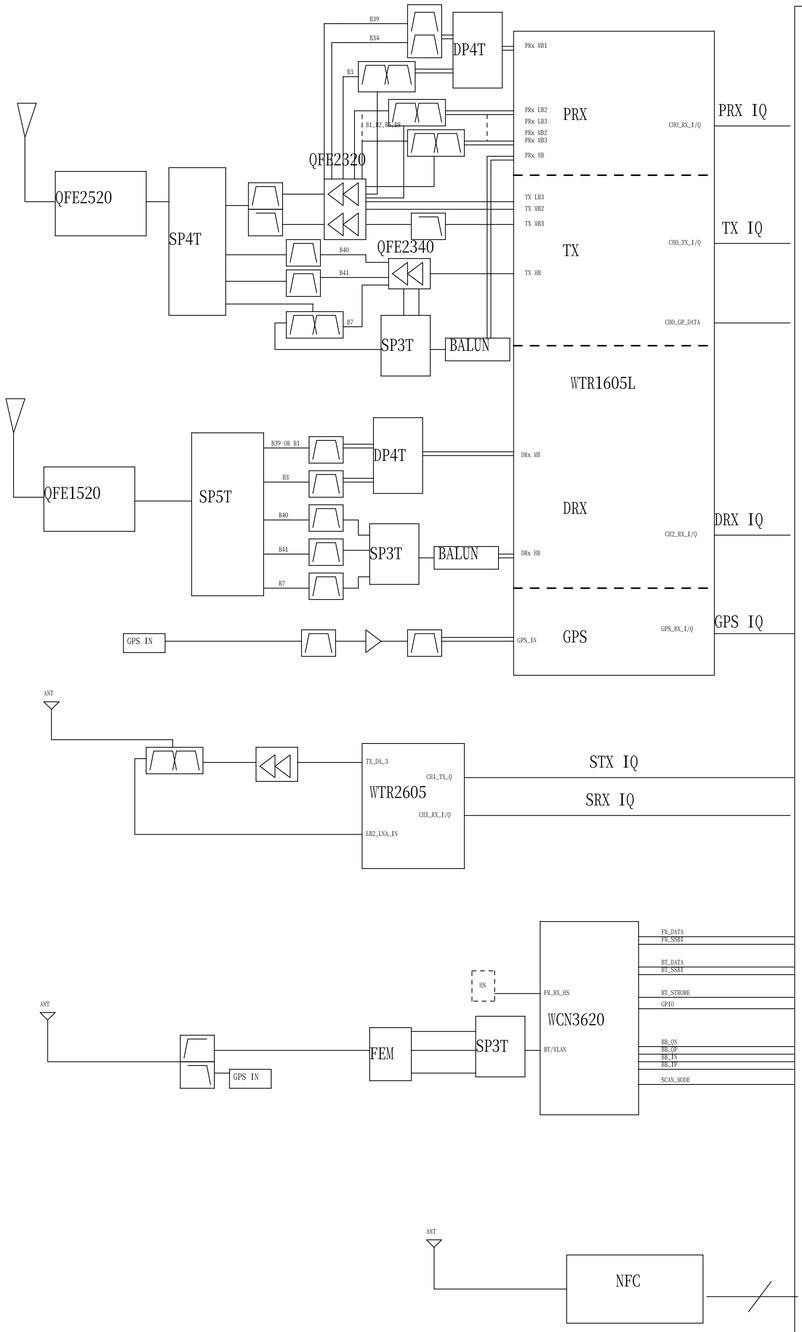

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MSM8916 GPIO Configuration For QRD8916

GPIO_0	NC	GPIO_41	WCSS_WLAN_DATA_1	GPIO_82	WTR1_SWITCH_CTL3
GPIO_1	NC	GPIO_42	WCSS_WLAN_DATA_0	GPIO_83	WTR1_GSM_PA_HB_EN
GPIO_2	NC	GPIO_43	WCSS_WLAN_SET	GPIO_84	WTR1_SWITCH_CTL0
GPIO_3	NC	GPIO_44	WCSS_WLAN_CLK	GPIO_85	DRX_MODE_SEL2
GPIO_4	UART_MSM_TX	GPIO_45	WCSS_FM_SSBI	GPIO_86	NC
GPIO_5	UART_MSM_RX	GPIO_46	WCSS_FM_SDI	GPIO_87	WTR0_PA_R1
GPIO_6	SENSOR_I2C_SDA	GPIO_47	WCSS_BT_CTRL	GPIO_88	WTR1_PA_R1
GPIO_7	SENSOR_I2C_SCL	GPIO_48	WCSS_BT_DAT_STB	GPIO_89	WTR1_PA_R0
GPIO_8	NC	GPIO_49	LCD_ID	GPIO_90	DRX_MODE_SEL0
GPIO_9	NC	GPIO_50	CTP_ID	GPIO_91	GRFC_17
GPIO_10	NC	GPIO_51	MCAM_ID	GPIO_92	WTR1_PA_BCO_EN
GPIO_11	NC	GPIO_52	NC	GPIO_93	DRX_MODE_SEL1
GPIO_12	TP_RST_N	GPIO_53	UIM2_DATA	GPIO_94	WTR0_PA_R0
GPIO_13	TP_INT_N	GPIO_54	UIM2_CLK	GPIO_95	WTR1_SWITCH_CTRL2
GPIO_14	BUCK_SMB_I2C_SDA	GPIO_55	UIM2_RESET	GPIO_96	EXT_GPS_LNA_EN
GPIO_15	BUCK_SMB_I2C_SCL	GPIO_56	NC	GPIO_97	NC
GPIO_16	NC	GPIO_57	UIM1_DATA	GPIO_98	NC
GPIO_17	NC	GPIO_58	UIM1_CLK	GPIO_99	CH0_GSM_TX_PHASE_D0
GPIO_18	TP_I2C_SDA	GPIO_59	UIM1_RESET	GPIO_100	CH0_GSM_TX_PHASE_D1
GPIO_19	TP_I2C_SCL	GPIO_60	NC	GPIO_101	CH1_GSM_TX_PHASE_D0
GPIO_20	NFC_DISABLE	GPIO_61	UIM_BATT_ALM	GPIO_102	CH1_GSM_TX_PHASE_D1
GPIO_21	NFC_INT_N	GPIO_62	SMB_INT	GPIO_103	CH0_SSBI_WTR_RX
GPIO_22	NFC_I2C_SDA	GPIO_63	CDC_PDM_CLK	GPIO_104	CH0_SSBI_WTR_TX
GPIO_23	NFC_I2C_SCL	GPIO_64	CDC_PDM_SYNC	GPIO_105	CH1_SSBI_WTR_RX
GPIO_24	LCD_TE0	GPIO_65	CDC_PDM_TX	GPIO_106	CH1_SSBI_WTR_TX
GPIO_25	LCD_RST_N	GPIO_66	CDC_PDM_RX0	GPIO_107	KEY_VOL_UP_N
GPIO_26	CAM0_MCLK	GPIO_67	CDC_PDM_RX1	GPIO_108	NC
GPIO_27	CAM1_MCLK	GPIO_68	CDC_PDM_RX2	GPIO_109	NC
GPIO_28	SCAM_RST_N	GPIO_69	NC	GPIO_110	USB_HS_ID
GPIO_29	CAM_I2C_SDA	GPIO_70	RFFE1_CLK	GPIO_111	NC
GPIO_30	CAM_I2C_SCL	GPIO_71	RFFE1_DATA	GPIO_112	ACCL_INT1_N
GPIO_31	CAM_FLASH_TORCH_EN	GPIO_72	RFFE2_CLK	GPIO_113	ALSP_INT_N
GPIO_32	CAM_FLASH_TORCH_MODE	GPIO_73	RFFE2_DATA	GPIO_114	ACCL_INT2_N
GPIO_33	SCAM_PWDN	GPIO_74	NC	GPIO_115	GYRO_INT_N
GPIO_34	MCAM_PWDN	GPIO_75	GRFC_1	GPIO_116	NC
GPIO_35	MCAM_RST_N	GPIO_76	GRFC_2	GPIO_117	NC
GPIO_36	NC	GPIO_77	WTR1_GSM_PA_LB_EN	GPIO_118	NC
GPIO_37	FORCE_USB_BOOT	GPIO_78	TDS_B34/B39_PA_ON	GPIO_119	CTP_LED_EN
GPIO_38	NC	GPIO_79	WTR1_SWITCH_CTL1	GPIO_120	CAM_VCM_PWDN
GPIO_39	WCSS_BT_SSBI	GPIO_80	WTR0_RX_ON	GPIO_121	NC
GPIO_40	WCSS_WLAN_DATA_2	GPIO_81	WTR0_RF_ON		

PM8916 GPIO/MPP Configuration For QRD8916

GPIO_1	UIM_BATT_ALM	MPP_1	VDD_PX_BIAS_MPP
GPIO_2	NFC_CLK_REQ	MPP_2	SCAM_ID
GPIO_3	NC	MPP_3	VREF_DAC_MPP
GPIO_4	NC	MPP_4	PWM_OUT

Sensors I2C Address List For QRD8916

I2C GPIO	Device	MPN	Write Address	Read Address
GPIO[6:7]	Accelerometer	BMA223	0x30	0x31
	Magnetometer	MMC34160PJ	0x60	0x61
	Gyroscope	MPU-6050C	0xD0	0xD1
	ALS/Proximity	TMD27723	0x72	0x73
GPIO[18:19]	Touch Screen	FT6346	0x70	0x71
GPIO[29:30]	Front Camera	OV2680	0x20	0x21
	Rear Camera	OV5648	0x6C	0x6D

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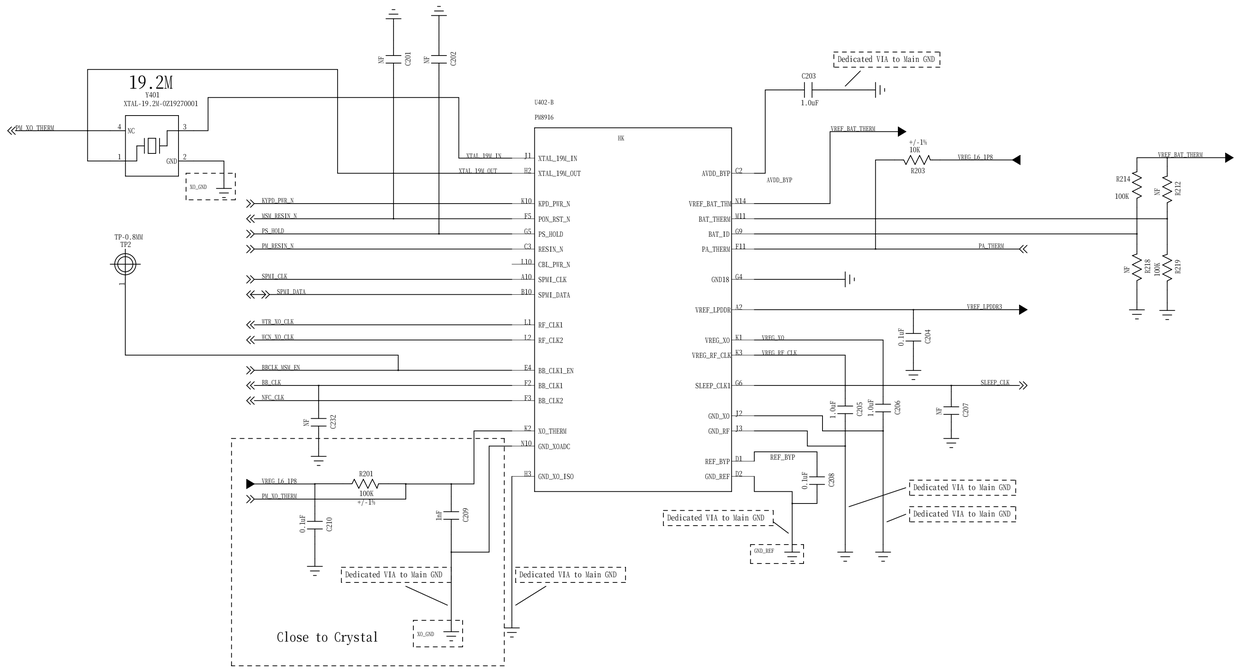

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Pins	Hi Z	GND
OPT_1	No SMB1360	SMB1360 present
OPT_2	No Ext APC buck (S5)	Ext APC buck (S5) present

PM8916 support +15V input protected

Dedicated sense trace to battery connector

Already have 47k resistor in subboard

Close to connector
if using PMIC's linear charger
C0301 should assemble 4.7uF/16V capacitor
R0307 should assebble 0 ohm
R0302 should NC
R0312 should be 0ohm
BAT_THERM switch to PM8916

SMB1360 can support +28V input protected

Software Caution:
Need software to config MSM
GPIO internal pullup

Dedicated differential sense trace from resistor

Dedicated differential sense trace to battery connector

NOTE1: Seperate the AGND from the PGND

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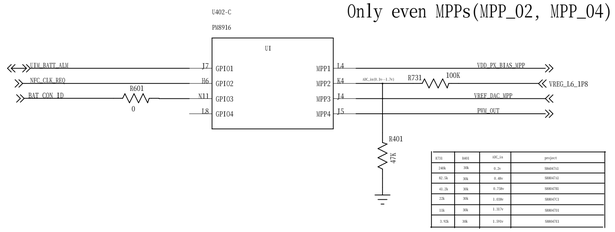
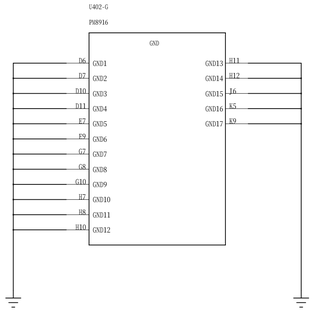
Only odd MPPs (MPP_01, MPP_03) can be configured as analog outputs.
Only even MPPs(MPP_02, MPP_04) have current sink capability.





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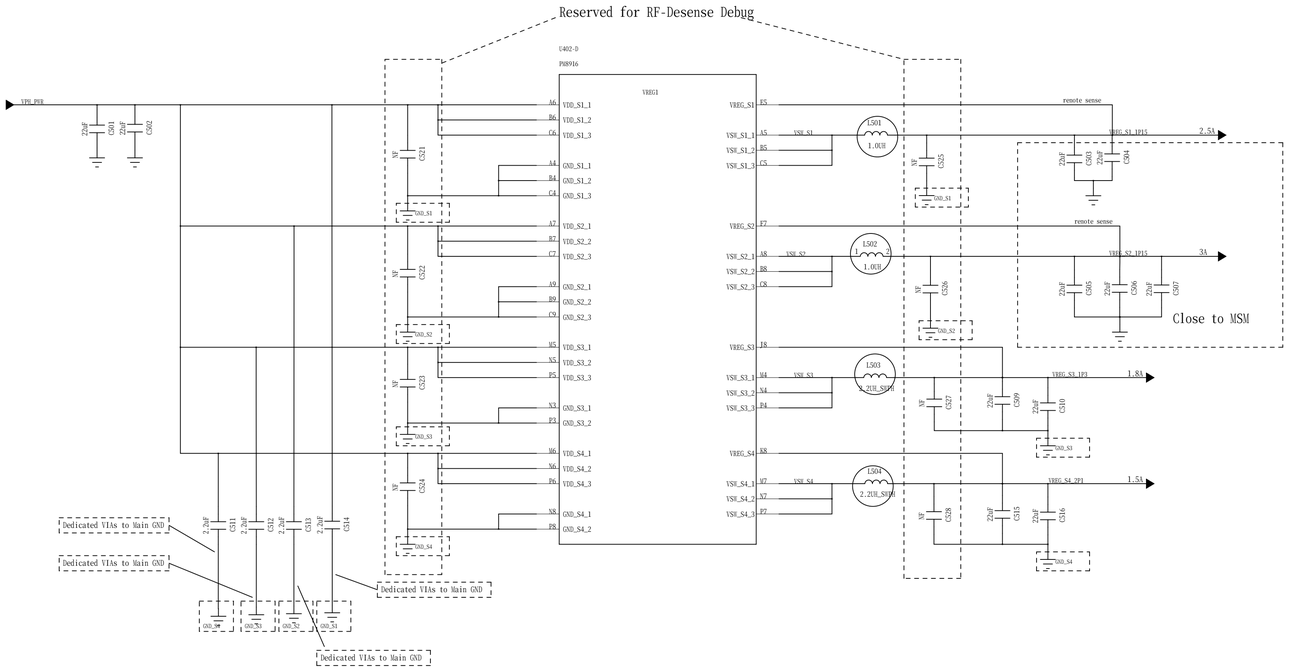

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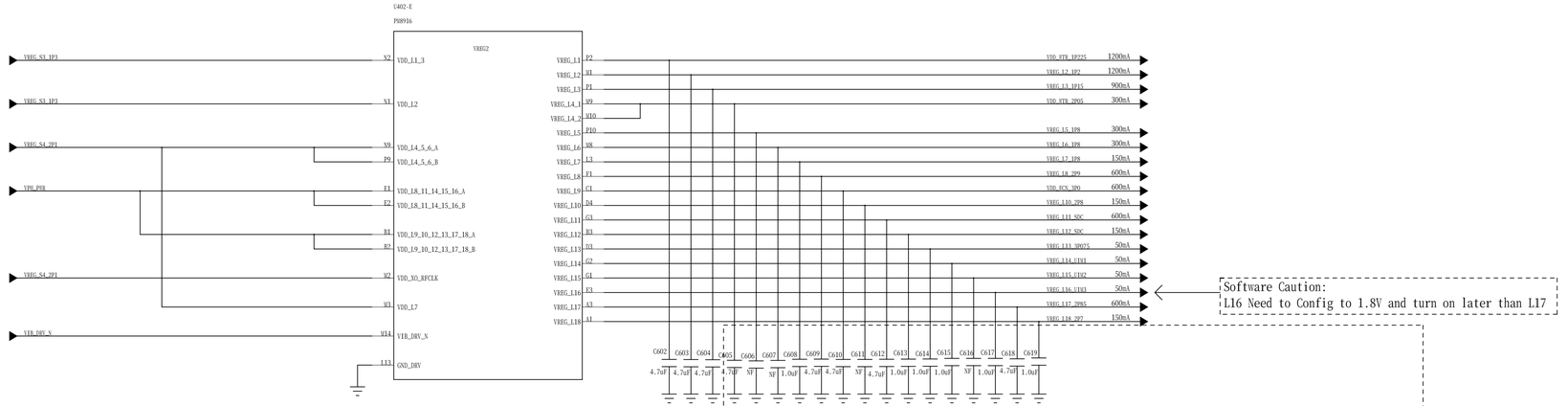

Software Caution:
L16 Need to Config to 1.8V and turn on later than L17

PSEUDO CAPLESS LDOs
L4, L5, L6, L7, L8, L9, L10, L11, L12, L13, L14, L15, L16, L17, L18

Local output capacitor can NC/Remove depends on remote load capacitors and PCB layout
LDOs L8, L9, L11, L17 each shall have total effective output capacitance of 4.7uF.
Please refer to PM8916 training slides(80-NK808-21) for limits on LDO output trace resistance and inductance
Rest of the LDOs shall each have total effective output capacitance of 1.0uF.

The Pseudo capless LDOs must comply to following tolerable routing requirement:
P50: R<400mOhms, L<45nH
P150: R<200mOhms, L<20nH
P300: R<150mOhms, L<18nH
P600: R<150mOhms, L<18nH

Please ensure when you DNI the output capacitor of Pseudo capless LDOs
the effective total capacitance at load side shall match above requirement

NC first

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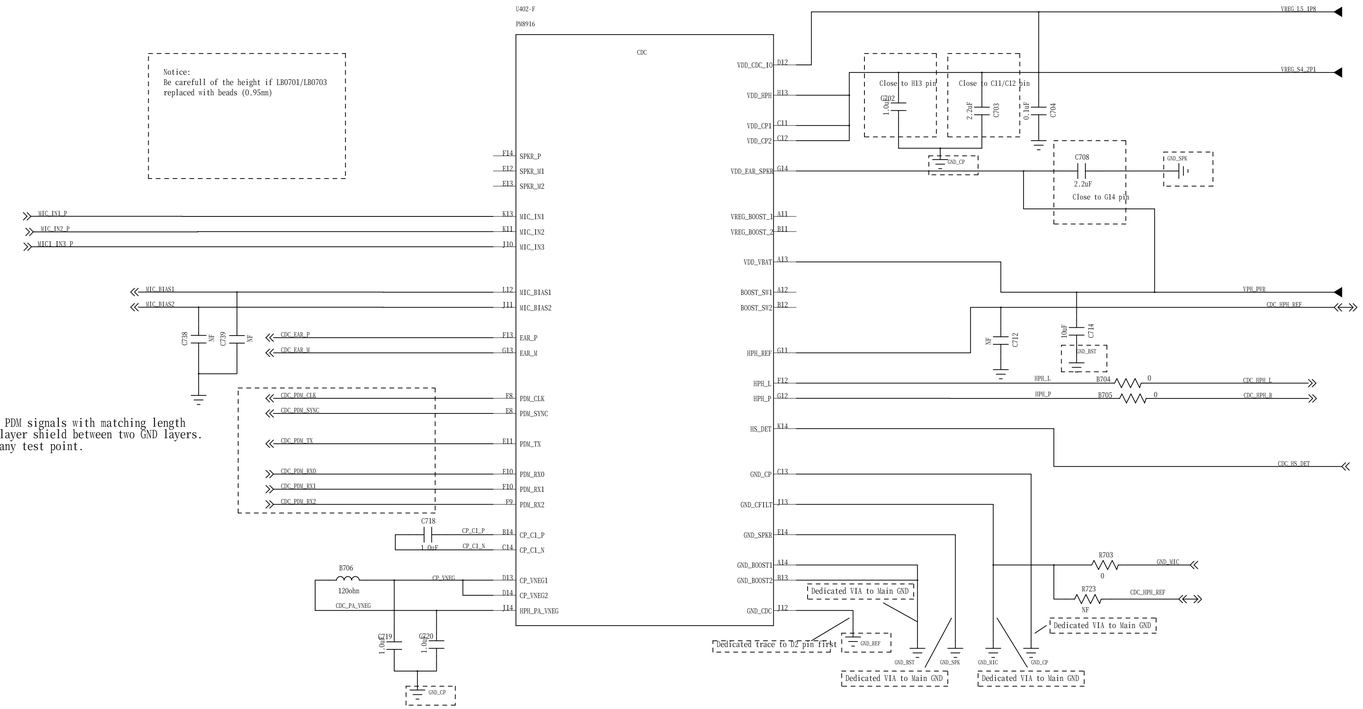

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For BBCLK1,AD35 will be the first load and AL34 will be the final load
Should be a daisy chain route

U401-A

MSM8916

CONTROL

	AR8	VLAN_XO
BB_CLK	AL34	CXO
BBCLK_MSM_EN	A36	CXO_EN
SLEEP_CLK	C36	SLEEP_CLK
MSM_RESETX_N	F35	RESET_N
MSM_RESOUT_N	AY1	RESOUT_N
	AU32	MODE_0
	AT31	MODE_1
	E36	PS_HOLD
JTAG_RESOUT_N	K1	SRST_N
JTAG_TCK	H1	TCK
JTAG_TDI	H3	TDI
JTAG_TDO	J2	TDO
JTAG_TMS	L2	TMS
JTAG_TRST_N	K3	TRST_N
VREF_LPDDR3	J12	EBI_VREF_CA
	H21	EBI_VREF_D0_D2
VREF_LPDDR3	J26	EBI_VREF_D1_D3
	AN8	DNC1
	BA36	DNC2
	AY35	DNC3

TP8
TP-0.80H

TP26
TP-0.80H

Star routing from C0803 to PINs
NC others CAPs first

VREF_LPDDR3

NF C0801

NF C0802

0.1uF C0803

SDC1_CLK	AE36	SDC1_CLK
SDC1_CMD	AH35	SDC1_CMD
SDC1_DATA_0	AD37	SDC1_DATA_0
SDC1_DATA_1	AF38	SDC1_DATA_1
SDC1_DATA_2	AG34	SDC1_DATA_2
SDC1_DATA_3	AK37	SDC1_DATA_3
SDC1_DATA_4	AG40	SDC1_DATA_4
SDC1_DATA_5	AG38	SDC1_DATA_5
SDC1_DATA_6	AF39	SDC1_DATA_6
SDC1_DATA_7	AF35	SDC1_DATA_7
SDC2_CLK	R4	SDC2_SDCARD_CLK
SDC2_CMD	N6	SDC2_SDCARD_CMD
SDC2_DATA_0	F3	SDC2_SDCARD_D0
SDC2_DATA_1	R6	SDC2_SDCARD_D1
SDC2_DATA_2	T7	SDC2_SDCARD_D2
SDC2_DATA_3	P7	SDC2_SDCARD_D3

R802
0

R804
0

Close to MSM

USB_HS_DP	AC40	USB_HS_D_P
USB_HS_DM	AR39	USB_HS_D_M
USB_HS_ID	AA40	
USB_HS_REXT	AC38	USB_HS_REXT
USB_HS_SYSCLOCK	AU35	BB_CLK

R803
200
+/-1%

PMIC_SPMI_CLK	AK37	SPMI_CLK
PMIC_SPMI_DATA	AJ36	SPMI_DATA

C804 NF

C805 NF

SPMI Data trace length should be within +/-2mm of SMPI Clock trace length. 33pF shunt caps must be no more than 2mm away from the MSM.

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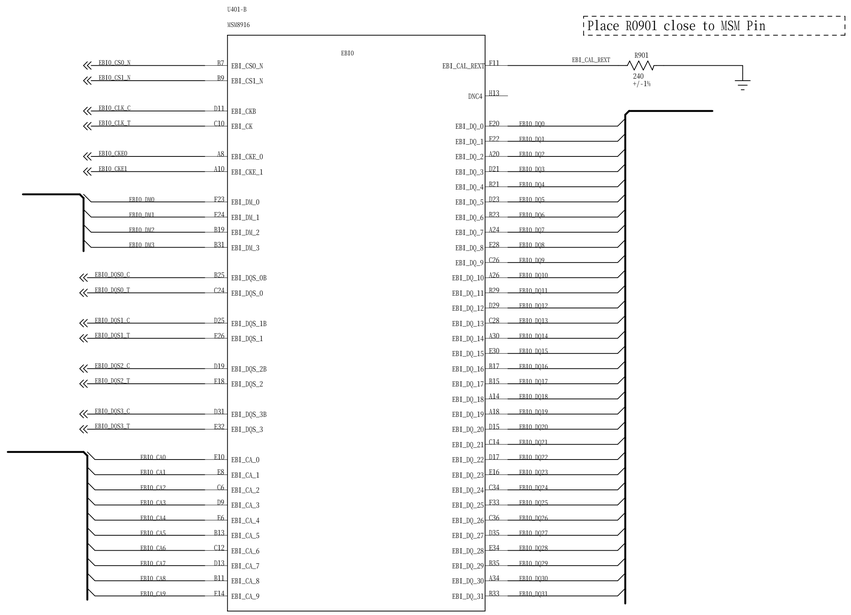

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U401-C

MSM8916

GPIO1

BAT_ID_1V8	BA38	GPIO_0	
	BB39	GPIO_1	*
	AY35	GPIO_2	
	AT37	GPIO_3	
UART_MSM_TX	AT9	GPIO_4	
UART_MSM_RX	AY1	GPIO_5	*
SENSOR_I2C_SDA	AY3	GPIO_6	
SENSOR_I2C_SCL	AY3	GPIO_7	
LCD_2PRS_EN	H5	GPIO_8	
	G2	GPIO_9	*
	H1	GPIO_10	
	E2	GPIO_11	*
TP_RST_N	AG39	GPIO_12	*
TP_INT_N	AG35	GPIO_13	*
BCCK_SMB_I2C_SDA	AV40	GPIO_14	
BCCK_SMB_I2C_SCL	AW36	GPIO_15	
	K7	GPIO_16	
	G6	GPIO_17	
TP_I2C_SDA	J6	GPIO_18	
TP_I2C_SCL	F4	GPIO_19	
NFC_DISABLE	AT7	GPIO_20	*
NFC_INT_N	AY6	GPIO_21	*
FED_I2C_SDA	AU7	GPIO_22	
LED_I2C_SCL	BA12	GPIO_23	
LCD_TE0	AT5	GPIO_24	
LCD_RST_N	AU4	GPIO_25	*
CAM0_MCLK	B3	GPIO_26	
CAM1_MCLK	C2	GPIO_27	
SCAM_RST_N	F1	GPIO_28	*
CAM_I2C_SDA	B3	GPIO_29	
CAM_I2C_SCL	F3	GPIO_30	
CAM_FLASH_TORCH_EN	D3	GPIO_31	*
CAM_FLASH_TORCH_MODE	D1	GPIO_32	
SCAM_PWDN	G4	GPIO_33	
MCAM_PWDN	F5	GPIO_34	*
MCAM_RST_N	A4	GPIO_35	*
	C4	GPIO_36	*

GPIO1

	E4	GPIO_37	*
	R37	GPIO_38	*
	BC2	GPIO_39	
	BB5	GPIO_40	
	BB1	GPIO_41	
	BD3	GPIO_42	
	B16	GPIO_43	
	BB3	GPIO_44	
	AY5	GPIO_45	
	AY9	GPIO_46	
	B14	GPIO_47	
	AY8	GPIO_48	
	V37	GPIO_49	*
	AA34	GPIO_50	*
	V35	GPIO_51	*
	AA38	GPIO_52	
	L40	GPIO_53	
	K39	GPIO_54	*
	U39	GPIO_55	
	AA36	GPIO_56	
	G40	GPIO_57	
	J40	GPIO_58	
	L38	GPIO_59	
	Y39	GPIO_60	
	AL38	GPIO_61	
	G34	GPIO_62	*
	AJ40	GPIO_63	
	AB39	GPIO_64	
	AE39	GPIO_65	
	AJ38	GPIO_66	*
	AK35	GPIO_67	
	AL40	GPIO_68	*
	L36	GPIO_69	*
	AN38	GPIO_70	
	AR38	GPIO_71	
	AT39	GPIO_72	
	AU40	GPIO_73	

FORCE_USB_BOOT	
WCNSS_BT_SSBI	
WCNSS_HIAVN_DATA_2	
WCNSS_HIAVN_DATA_1	
WCNSS_HIAVN_DATA_0	
WCNSS_HIAVN_SET	
WCNSS_HIAVN_CLK	
WCNSS_FM_SSBI	
WCNSS_FM_SDI	
WCNSS_BT_CTRL	
WCNSS_BT_DAT_STR	
LCD_ID	
CTP_ID	
WCAM_ID	
UIM2_DATA	
UIM2_CLK	
UIM2_RESET	
UIM2_PRESENT	
UIM1_DATA	
UIM1_CLK	
UIM1_RESET	
UIM1_PRESENT	
UIM_BATT_ALM	
NFC_IRQ	Need software to config internal pullup
NFC_ESE_PWR_REQ	
CDC_PDM_SYNC	
CDC_PDM_TX	
CDC_PDM_RX0	
CDC_PDM_RX1	
CDC_PDM_RX2	
RFFE1_CLK	
RFFE1_DATA	
RFFE2_CLK	
RFFE2_DATA	

For RF De-sense Debug

R1009 0 CAM0_MCLK_RB

R1010 0 CAM1_MCLK_RB

C806 33PF

C807 33PF

U401-D

MSM8916

GPIO2

LTE_BPI_OUT74	GRFC0	AP37	GPIO_74
ANT_TUNE_CTL	GRFC1	AV39	GPIO_75
ANT_TUNE_EN	GRFC2	AP35	GPIO_76
	GRFC3	BC4	GPIO_77
TDS_B34/B39_PA_ON	GRFC4	BE4	GPIO_78
LTE_BPI_OUT79	GRFC5	BC6	GPIO_79
WTR0_RX_ON	GRFC6	BG5	GPIO_80
WTR0_RF_ON	GRFC7	BG7	GPIO_81
LTE_BPI_OUT82	GRFC8	BC38	GPIO_82
	GRFC9	BC40	GPIO_83
LTE_BPI_OUT84	GRFC10	BC8	GPIO_84
B39_B3_SHITCH_SEL	GRFC11	BE6	GPIO_85
	GRFC12	BD39	GPIO_86
WTR0_PA_R1	GRFC13	BR37	GPIO_87
LTE_BPI_OUT88	GRFC14	BR35	GPIO_88
	GRFC15	BD35	GPIO_89
DRX_MODE_SEL0	GRFC16	BA32	GPIO_90
	GRFC17	BE38	GPIO_91
LTE_B7_PA_EN	GRFC18	AY33	GPIO_92
DRX_MODE_SEL1	GRFC19	BD33	GPIO_93
WTR0_PA_R0	GRFC20	BC34	GPIO_94
DRX_MODE_SEL2	GRFC21	BR31	GPIO_95
EXT_GPS_LNA_EN		BC32	GPIO_96
		C38	GPIO_97 *
		A38	GPIO_98 *
CH0_GSM_TX_PHASE_D0		BF34	GPIO_99
CH0_GSM_TX_PHASE_D1		BE36	GPIO_100
CH1_GSM_TX_PHASE_D0		BR33	GPIO_101
CH1_GSM_TX_PHASE_D1		BF32	GPIO_102
CH0_SSBI1_WTR_RX		AT35	GPIO_103
CH0_SSBI1_WTR_TX		AV40	GPIO_104
CH1_SSBI1_WTR_RX		AU36	GPIO_105
CH1_SSBI1_WTR_TX		AV39	GPIO_106
KEY_VOL_UP_N		H33	GPIO_107 *
		K35	GPIO_108 *
		J34	GPIO_109 *

VREG_L5_1P8

R1024 10K

Internal Use.

GPIO2

		B39	GPIO_110
		C40	GPIO_111 *
		AE36	GPIO_112 *
		D39	GPIO_113 *
		E40	GPIO_114 *
		E38	GPIO_115 *
		AR38	GPIO_116
		AV37	GPIO_117 *
		AE36	GPIO_118 *
		B40	GPIO_119
		F39	GPIO_120 *
		G38	GPIO_121

USB_HS_ID
GPIO_111 will be in O/P mode on Power On Reset, After Power On Reset, SW can configure it
I2S_2_DI
ALSP_INT_N
ACC1_INT2_N
GYRO_INT_N
I2S_2_WS
I2S_2_SCLK
I2S_2_DO
CAM_VCM_PWDN

VREG_L6_1P8

SENSORS

R1011 2.2K

R1012 2.2K

SENSOR_I2C_SDA

SENSOR_I2C_SCL

VREG_L6_1P8

CAMERAS

R1013 2.2K

R1014 2.2K

CAM_I2C_SDA

CAM_I2C_SCL

VREG_L5_1P8

EXT BUCK, SMB

R1015 2.2K

R1016 2.2K

BCCK_SMB_I2C_SDA

BCCK_SMB_I2C_SCL

VREG_L6_1P8

TOUCHSCREEN

R1017 2.2K

R1018 2.2K

TP_I2C_SDA

TP_I2C_SCL

VREG_L5_1P8

LED

R1019 2.2K

R1020 2.2K

LED_I2C_SDA

LED_I2C_SCL

I2C PULL-UP RESISTORS

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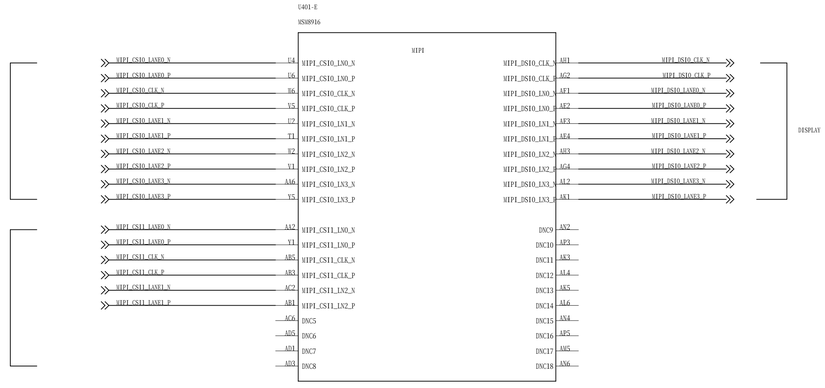

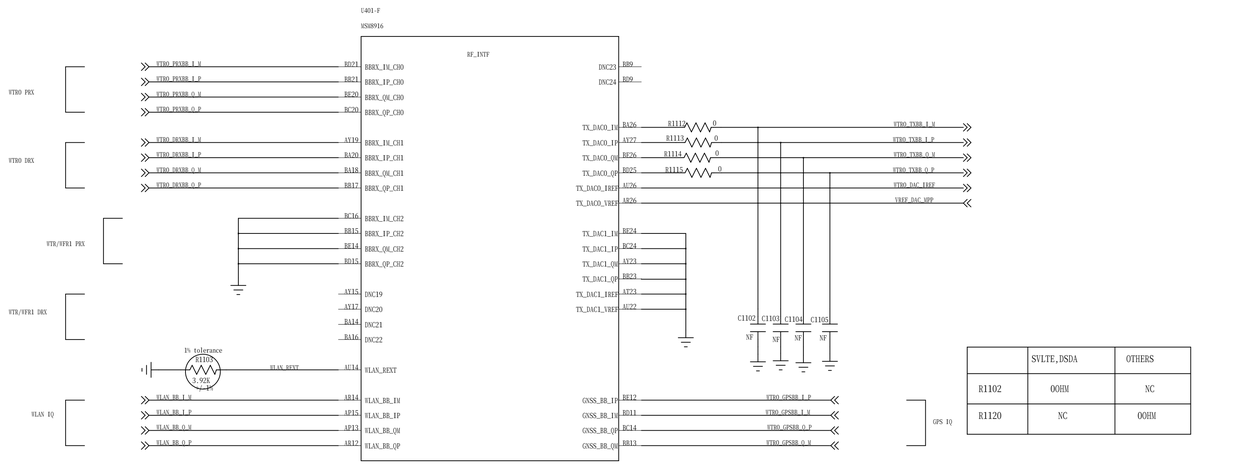

	SVLTE,DSDA	OTHERS
R1102	0OHM	NC
R1120	NC	0OHM

These NC resistor is for SVLTE or DSDA. Change then to 0ohm on other modes.

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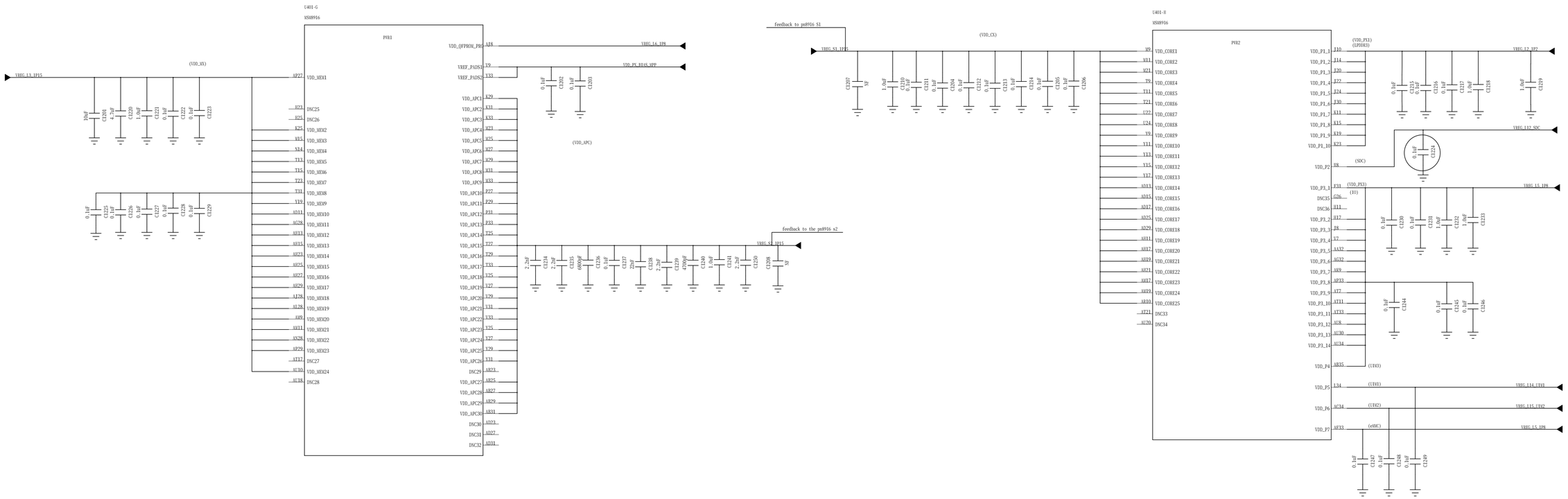

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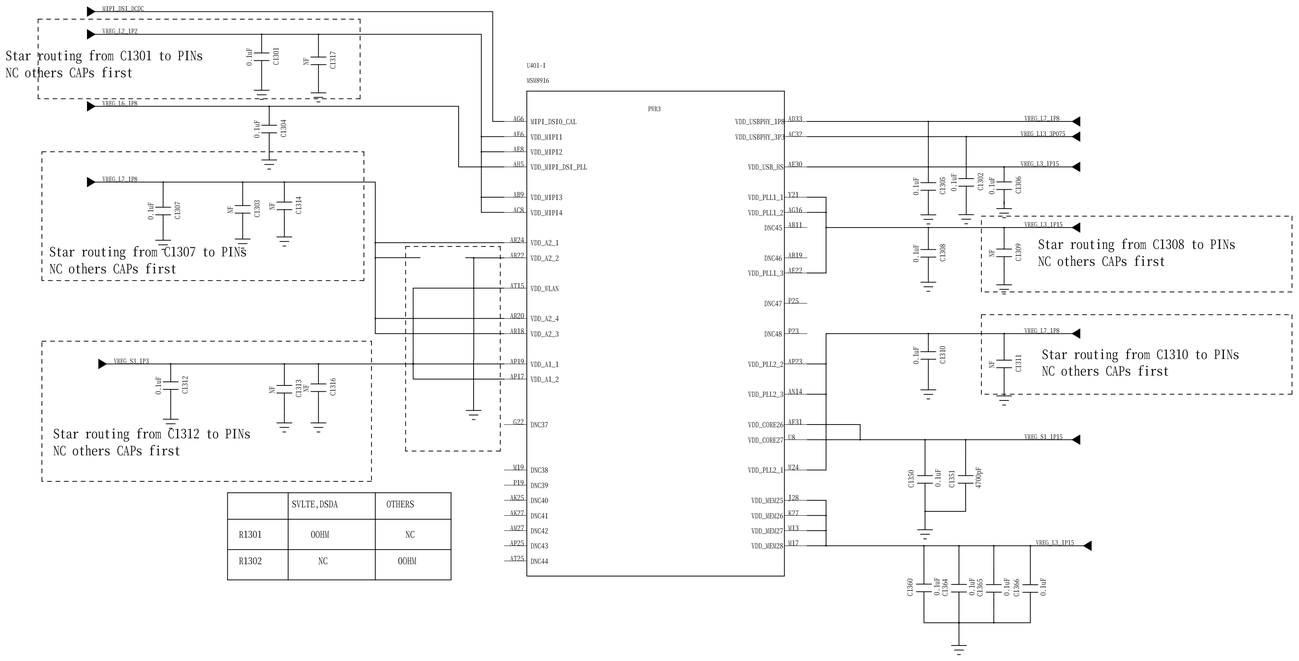

	SVLTE,DSDA	OTHERS
R1301	0OHM	NC
R1302	NC	0OHM

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U401-K

MSM8916

GND2

V35	GND101	GND139	AF21
V37	GND102	GND140	AF23
V39	GND103	GND141	AF29
R16	GND104	GND142	AH31
R26	GND105	GND143	AH33
R28	GND106	GND144	AL30
R30	GND107	GND145	AP25
R32	GND108	GND146	AP29
R34	GND109	GND147	AP31
R36	GND110	GND148	AP33
R38	GND111	GND149	AP9
R40	GND112	GND150	AR4
AA24	GND113	GND151	AR6
AA26	GND114	DNC45	AT1
AA28	GND115	DNC46	AT3
AA30	GND116	GND154	AT29
AB15	GND117	DNC47	AU2
AB17	GND118	GND156	AU6
AP9	GND119	GND157	AU12
AF11	GND120	GND158	AU38
AF13	GND121	GND159	AV5
AF15	GND122	GND160	AV9
AF17	GND123	GND161	AV11
AF19	GND124	GND162	AV17
AF21	GND125	GND163	AV19
AF23	GND126	GND164	AV33
AF25	GND127	GND165	AY2
AF27	GND128	GND166	AY4
AF29	GND129	GND167	AY10
AG20	GND130	GND168	AY34
AH9	GND131	GND169	AY11
AH31	GND132	GND170	BA8
AJ34	GND133	GND171	BB7
AK11	GND134	GND172	BC36
AK13	GND135	GND173	BD37
AK15	GND136	GND174	BD8
AK17	GND137	GND175	BE10
AK19	GND138		

U401-J

MSM8916

GND1

A6	GND1	GND51	AC37
A12	GND2	GND52	AC39
A16	GND3	GND53	AC2
A22	GND4	GND54	AC4
A28	GND5	GND55	AC8
A32	GND6	GND56	AC27
B5	GND7	GND57	AC26
C8	GND8	GND58	AC28
C18	GND9	GND59	AC30
C20	GND10	GND60	AC32
C32	GND11	GND61	AC34
D5	GND12	GND62	AC36
D7	GND13	GND63	AC38
D27	GND14	GND64	AC40
E12	GND15	GND65	AD5
E7	GND16	GND66	AD9
F19	GND17	GND67	AD11
F25	GND18	GND68	AD13
F27	GND19	GND69	AD15
F29	GND20	GND70	AD17
F37	GND21	GND71	AD35
G8	GND22	GND72	AD37
G10	GND23	GND73	AD39
G12	GND24	GND74	AE8
G14	GND25	GND75	AE26
G16	GND26	GND76	AE28
G18	GND27	GND77	AE30
G28	GND28	GND78	AE32
G30	GND29	GND79	AE34
G32	GND30	GND80	AE36
H35	GND31	GND81	AE38
H37	GND32	GND82	AE40
J16	GND33	GND83	AT17
J18	GND34	GND84	AT19
J32	GND35	GND85	AT35
J36	GND36	GND86	AT37
J38	GND37	GND87	AT39
K5	GND38	GND88	AU26
K9	GND39	GND89	AU28
K17	GND40	GND90	AU30
K21	GND41	GND91	AU32
K37	GND42	GND92	AU34
L4	GND43	GND93	AU36
L6	GND44	GND94	AU38
L8	GND45	GND95	AU40
L30	GND46	GND96	AV11
L32	GND47	GND97	AV13
M5	GND48	GND98	AV15
M7	GND49	GND99	AV17
M35	GND50	GND100	AV19

U401-L

MSM8916

GND3

AJ4	GND176	GND216	AC30
AF5	GND177	DNC48	F21
AU7	GND178	GND218	G24
AJ2	GND179	GND219	G20
AK7	GND180	GND220	AN20
AT8	GND181	GND221	AB13
AU1	GND182	GND222	Y23
AW3	GND183	GND223	Y21
T3	GND184	GND224	Y23
T5	GND185	GND225	Y22
V3	GND186	GND226	AG18
W4	GND187	GND227	AF16
Y3	GND188	GND228	AR21
AA4	GND189	GND229	AD19
AA8	GND190	GND230	AF20
AC4	GND191	GND231	AD21
AD7	GND192	GND232	AJ24
AT27	GND193	GND233	AJ24
AU28	GND194	GND234	F21
BA28	GND195	GND235	E22
BB27	GND196	GND236	AW23
BC26	GND197	GND237	AV21
BC28	GND198	GND238	AU15
BD27	GND199	GND239	AU13
BE28	GND200	GND240	AV15
BR25	GND201	GND241	AV14
BA22	GND202	GND242	AY13
BC22	GND203	GND243	BA12
BD23	GND204	GND244	BC12
BF22	GND205	GND245	BD13
AT13	GND206		
AV13	GND207		
AE12	GND208		
AY21	GND209		
BD19	GND210		
BE18	GND211		
BB19	GND212		
BC18	GND213		
BD17	GND214		
BE16	GND215		

MIPI_DSI_LDO is configured to be LDO mode defaultly.
if using DC/DC mode,
L1401 need to be replaced with 2.2uH inductor

MIPI_DSI_DCDC

L1401
0_NF

C1401
2.2uF

CAD NOTE: Loop to be shorted with MSM AB7 pin

U401-H

MSM8916

NC_DNC

A2	NC1	NC24	AY27
A40	NC2	NC25	AY29
B1	NC3	NC26	AY31
D33	NC4	NC27	AY16
D37	NC5	NC28	AY18
F15	NC6	NC29	AY20
F17	NC7	NC30	AY22
H7	NC8	NC31	AY24
H15	NC9	NC32	AY26
H19	NC10	NC33	AY28
H27	NC11	NC34	AY30
H29	NC12	NC35	AY32
H31	NC13	NC36	AY25
AB7	MIPI_DSI_LDO	NC37	AY29
AJ16	NC15	NC38	BA24
AJ7	NC16	NC39	BA30
AP2	NC17	NC40	BB29
AT19	NC18	NC41	BC30
AU16	NC19	NC42	BD1
AU24	NC20	NC43	BD29
AV21	NC21	NC44	BF2
AV23	NC22	NC45	BF40
AV25	NC23		

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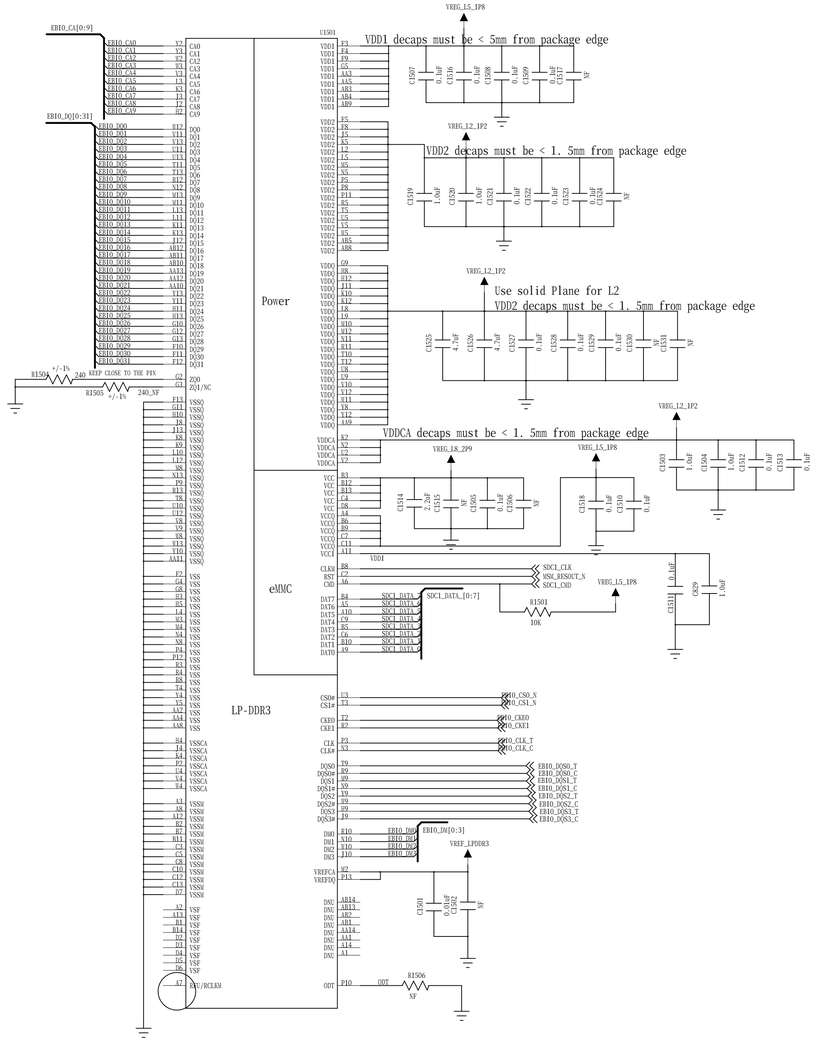
	MICRO/HYNIX	SAMSUNG
U1501.G3	ZQ1	NC
U1501.A7	RFU	RCLK



Note: Pull-up resistors on SDC1_DATA are PCB and eMMC vender dependent

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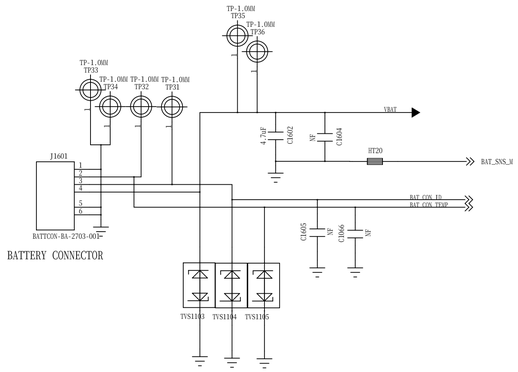

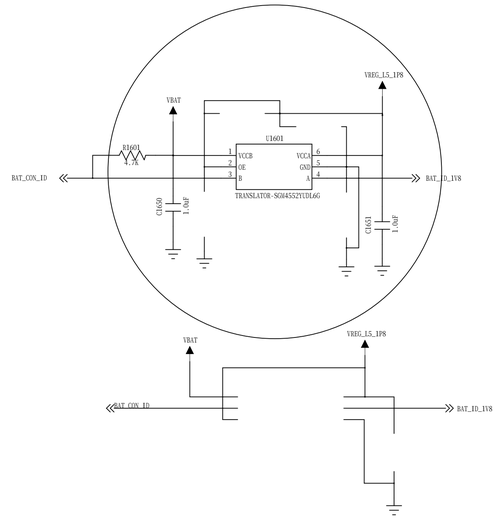

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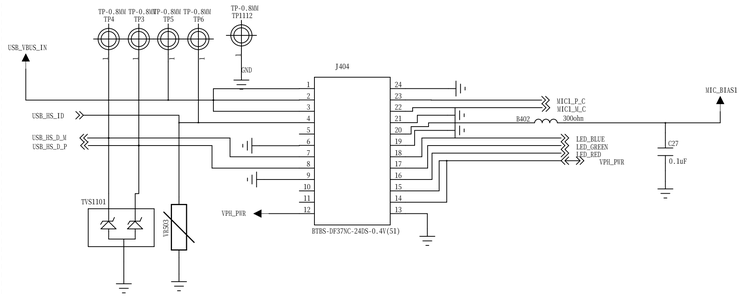
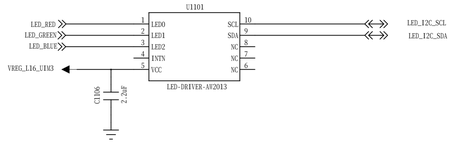
Sub Board Connector



MAIN MIC+USB+RGB



RGB

I2C Address	
Write	8B0x8A
Read	8B0x8B

Power on sequence: can be applied in any order

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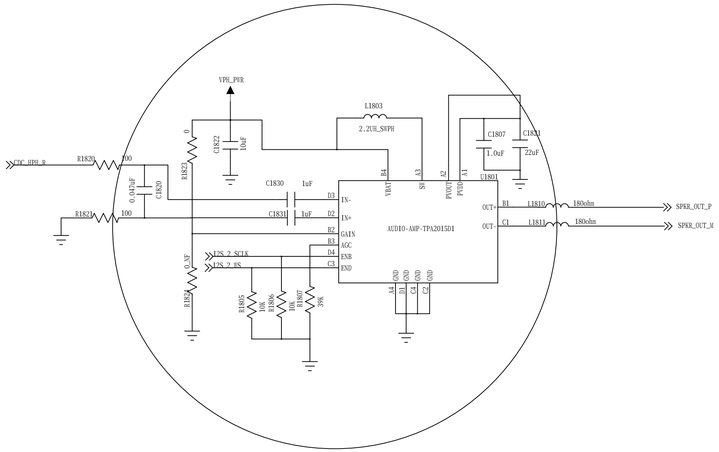
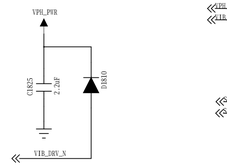
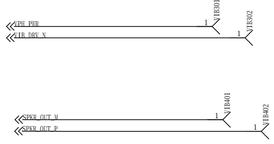
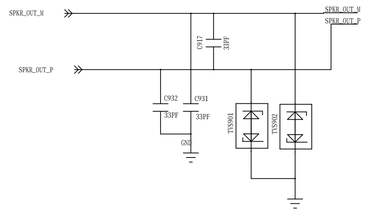
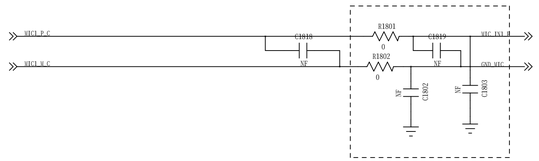




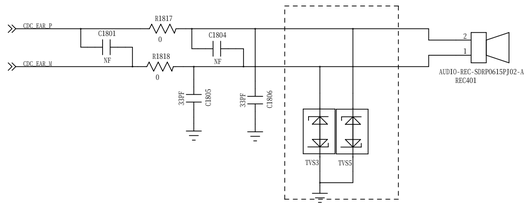




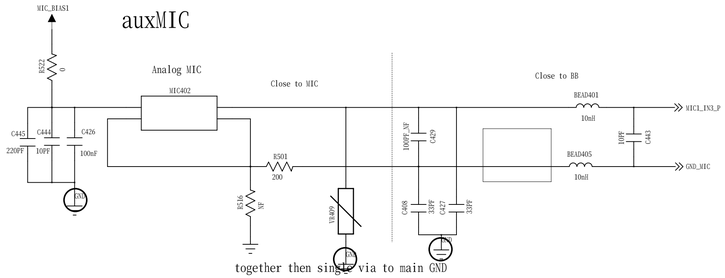
PRIMARY MIC



RECEIVER



auxMIC



together then single via to main GND

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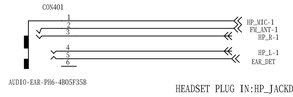
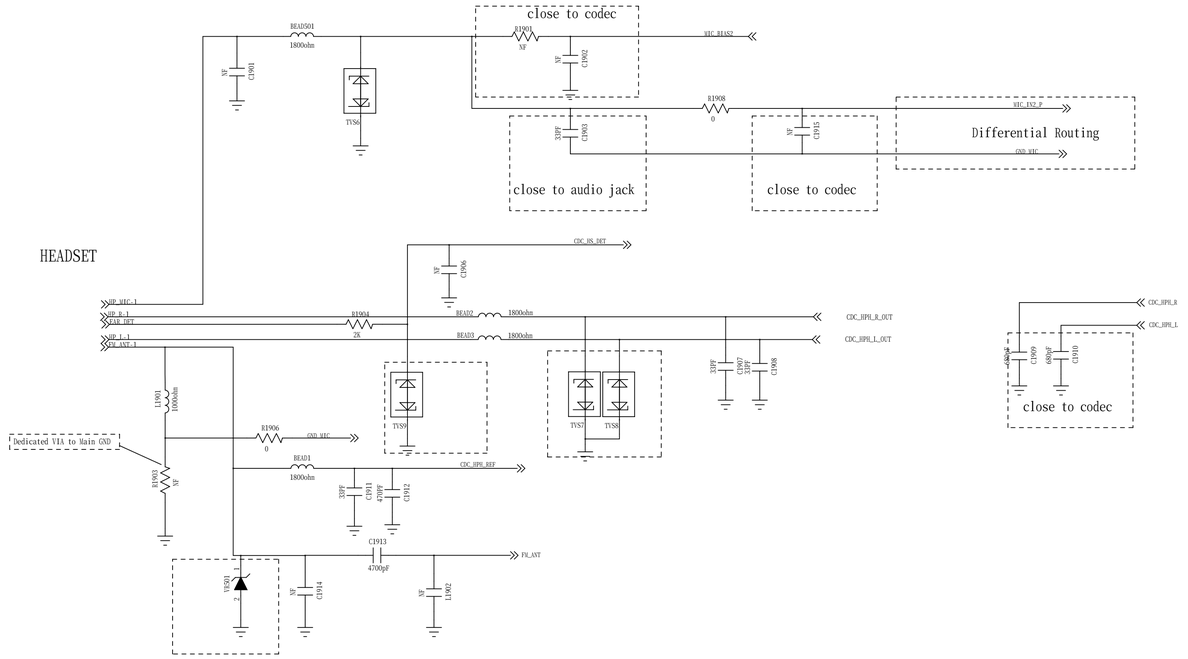
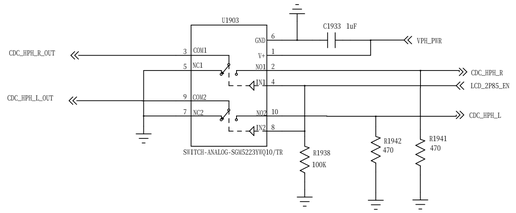


HEADSET PLUG IN:HP_JACKDET "0"
HEADSET PLUG OUT:HP_JACKDET "1"





IN=0, COM Connected to NC;
IN=1, COM Connected to NO;

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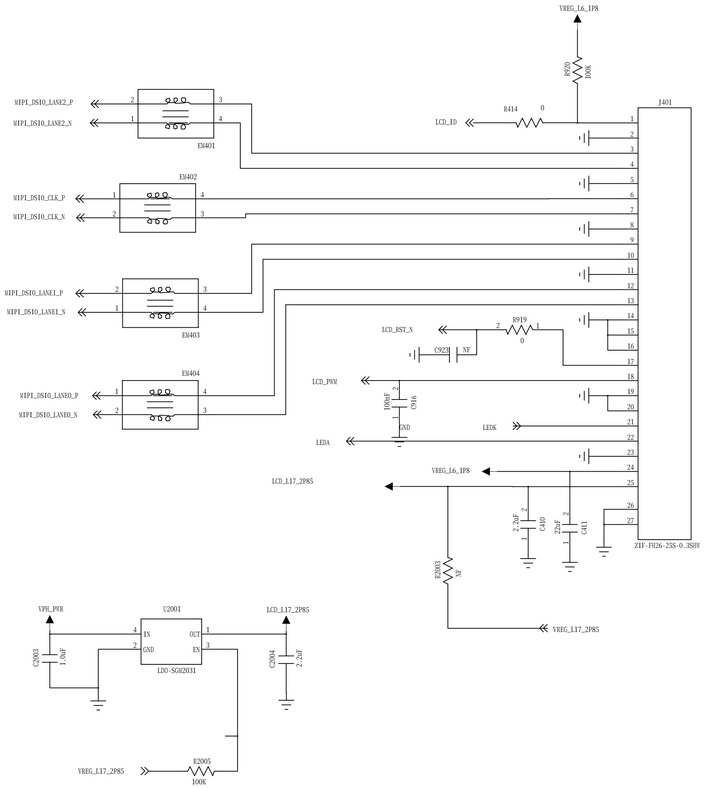
LCD

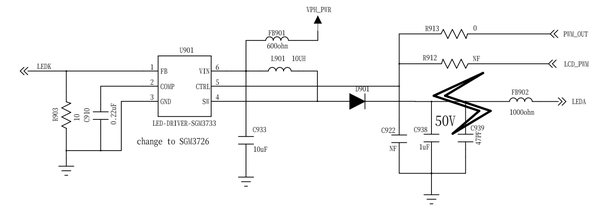
5.0' FWVGA

Module: KTF-050-D54980F

Driver: OTM8019A

Power on sequence: L6(IOVCC)/L17(VCC) can be applied in any order





Backlight LED Driver

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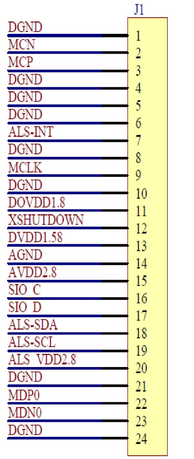
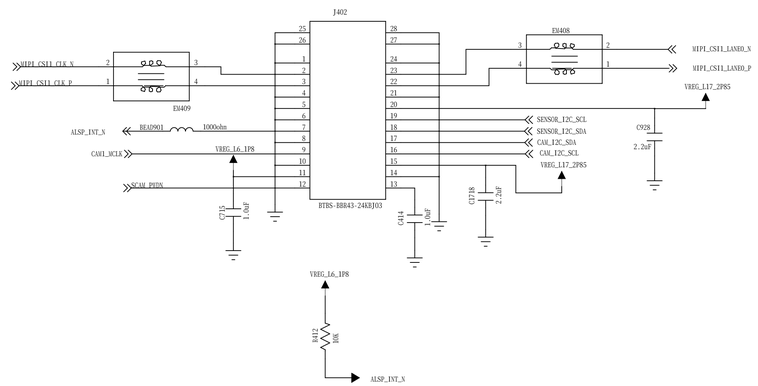
Sub Camera

FRONT_camera+LP Sensor+SPK



BBR13-24K6J01

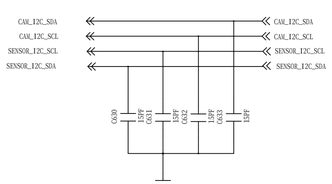


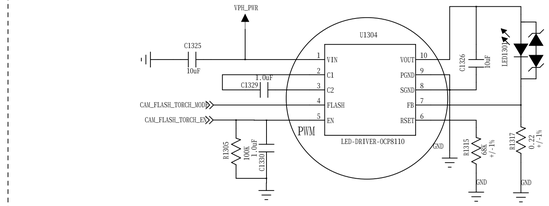
Main Camera / Sub Camera share power domain design
should double check the voltage level is compatible

light sensor 0x90 Write(STK3171)
light sensor 0x46 Write (LTR-558ALS)

Front Camera: 2M Module: Sensor: OV2680	I2C Address	
	Write	0x20
	Read	0x21

Power on sequence: L6(DOVDD)/L17(AVDD) can be applied in any order



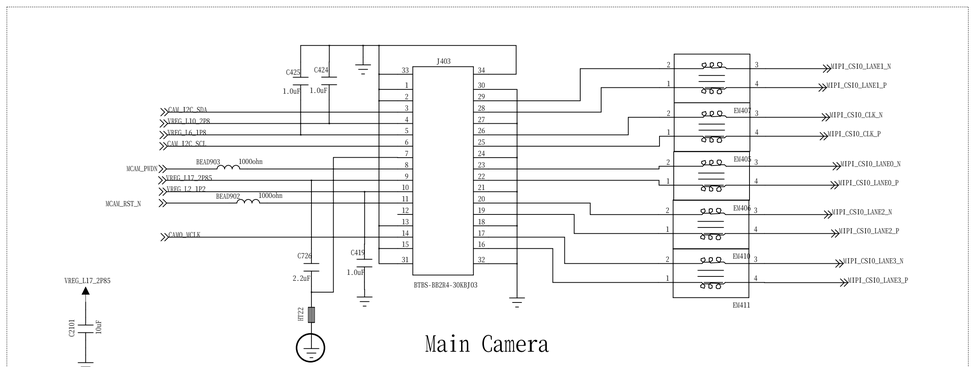


OCP8111 R1315=68K R1317=0.22
OCP8110 R1315=82K R1317=0.47

FLASH LED

light sensor 0x90 Write(STK3171)
light sensor 0x46 Write (LTR-558ALS)

Main Camera / Sub Camera share power domain design
should double check the voltage level is compatible



Main Camera

Rear Camera: 8Mp Module: Sensor: OV8865	I2C Address	
	Write	0x6C
	Read	0x6D

Power on sequence: L6(DOVDD) must before L17(AVDD)

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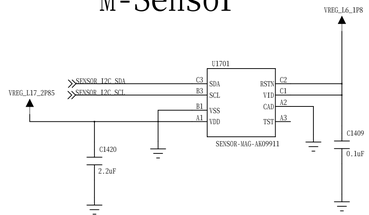
M-Sensor



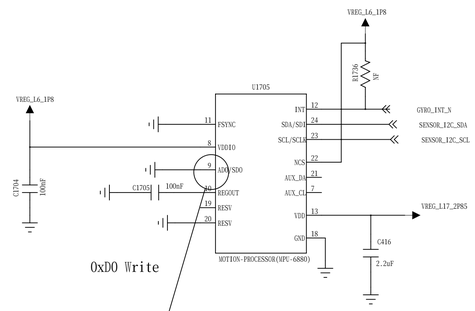
AKM: AKO9911

I2C Address@CAD=0	
Write	0x18
Read	0x19

I2C Address@CAD=1	
Write	0x1A
Read	0x1B

Power on sequence: L6/L17 can be applied in any order

Gyroscope



0xD0 Write

AD0=0 7b1101000 0x68

AD0=1 7b1101001 0x69

InvenSense: MPU6880

I2C Address@AD0=0	
Write	0xD0
Read	0xD1

I2C Address@AD0=1	
Write	0xD2
Read	0xD3

Power on sequence: L6/L17 can be applied in any order

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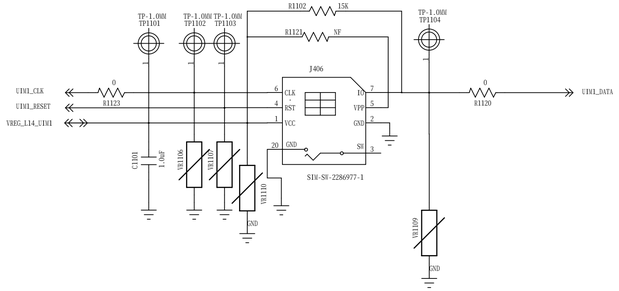
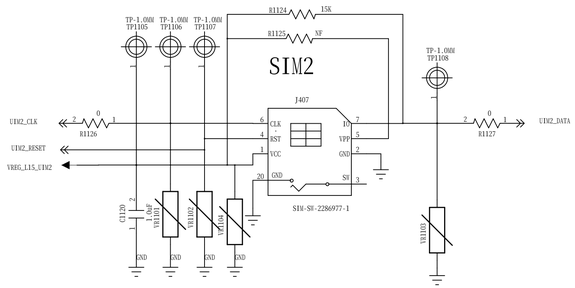
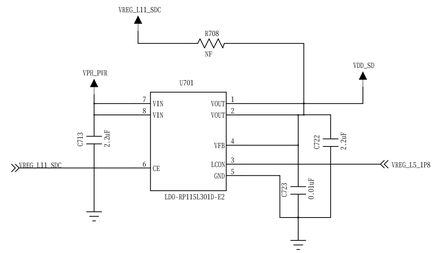
SIM1

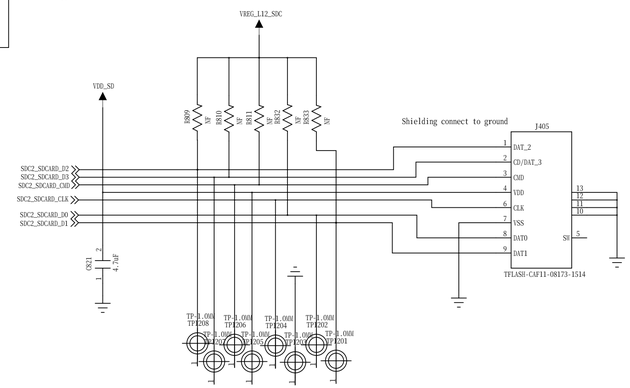


SIM2





SD CARD



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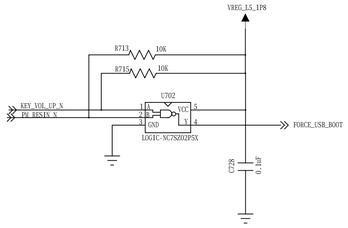

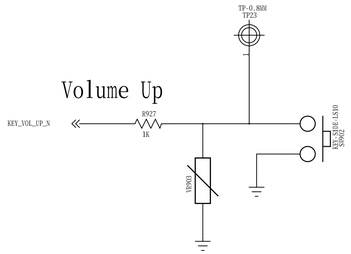

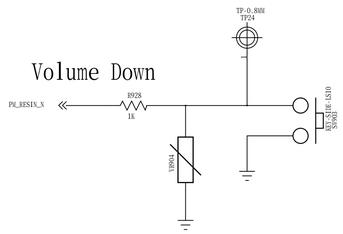

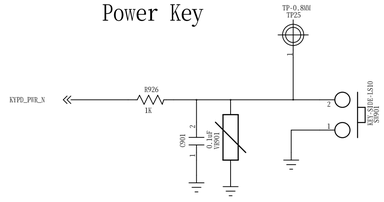

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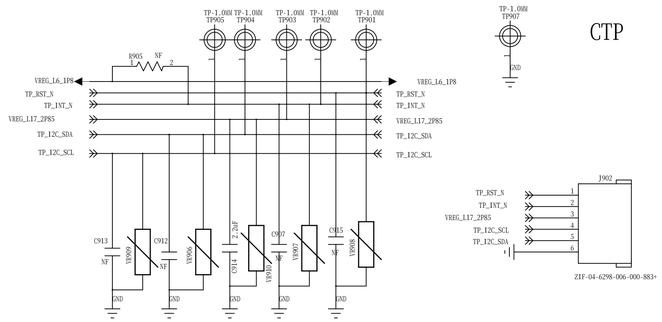

CTP

Power on sequence: Single Supply

FocalTech: FT5336	
I2C Address	
Write	0x70 It can be changed by FACTORY.
Read	0x71

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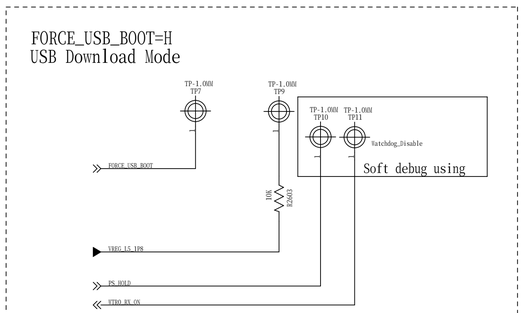

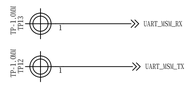

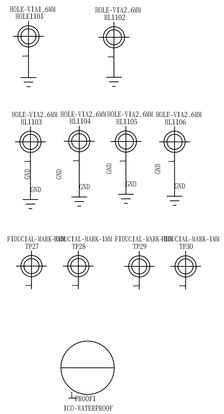

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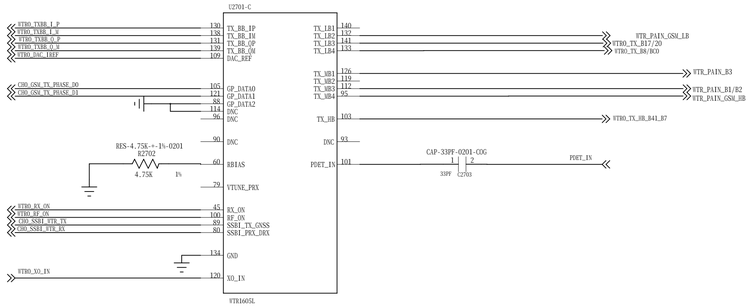
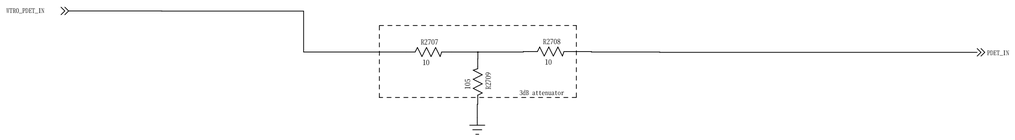


use C2701 for both SPDT power supply, and use C2705 for both SPDT control line



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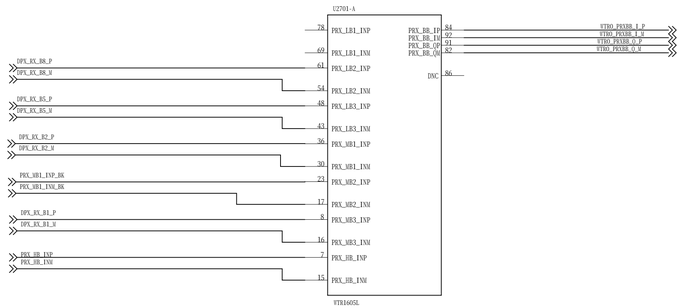

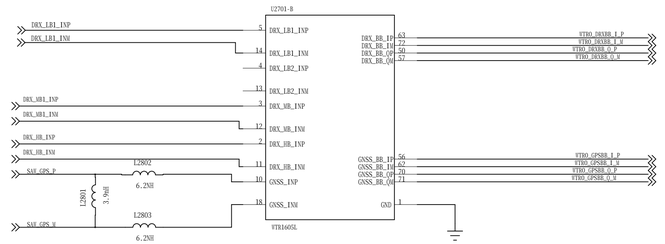

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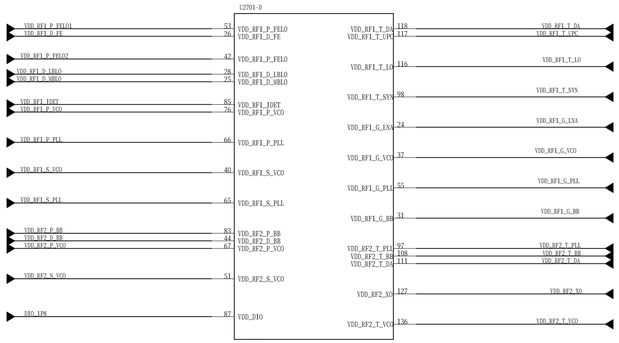

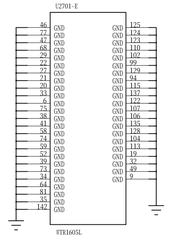

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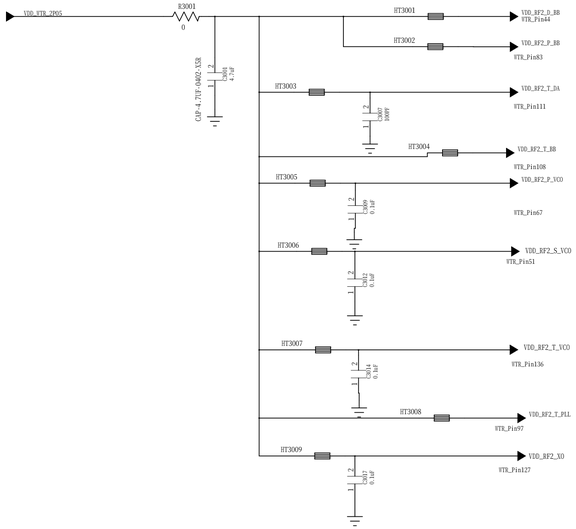

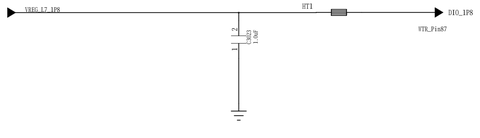

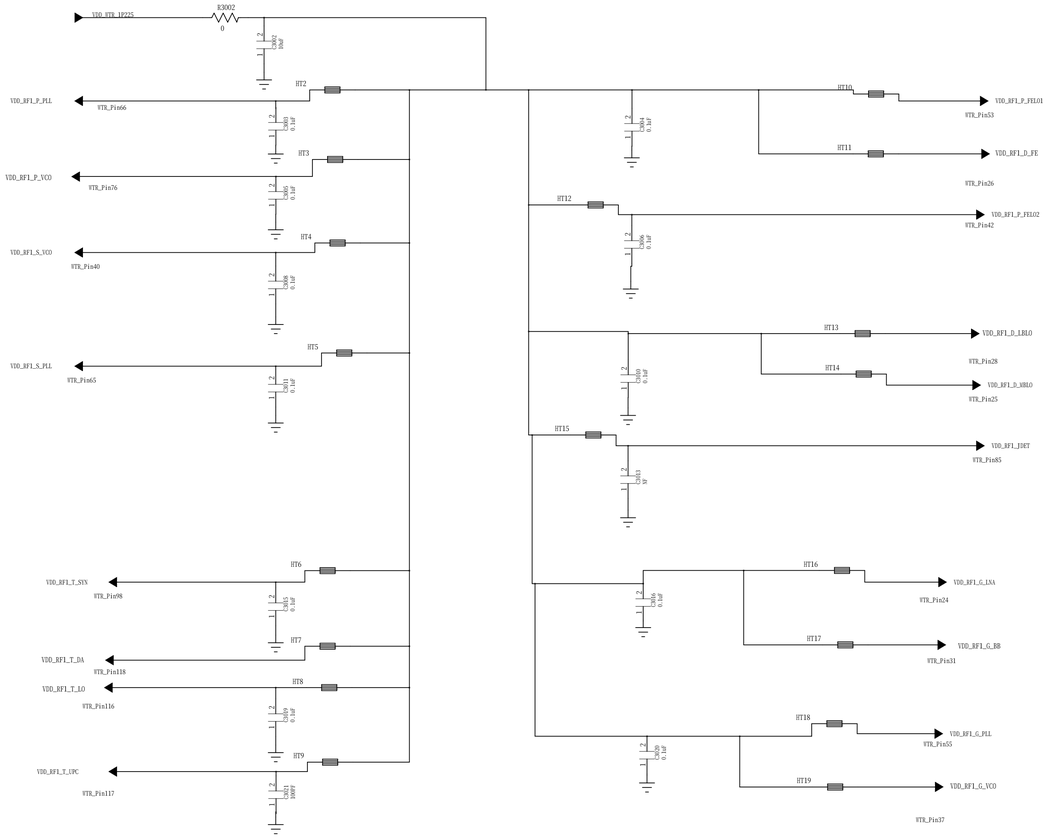

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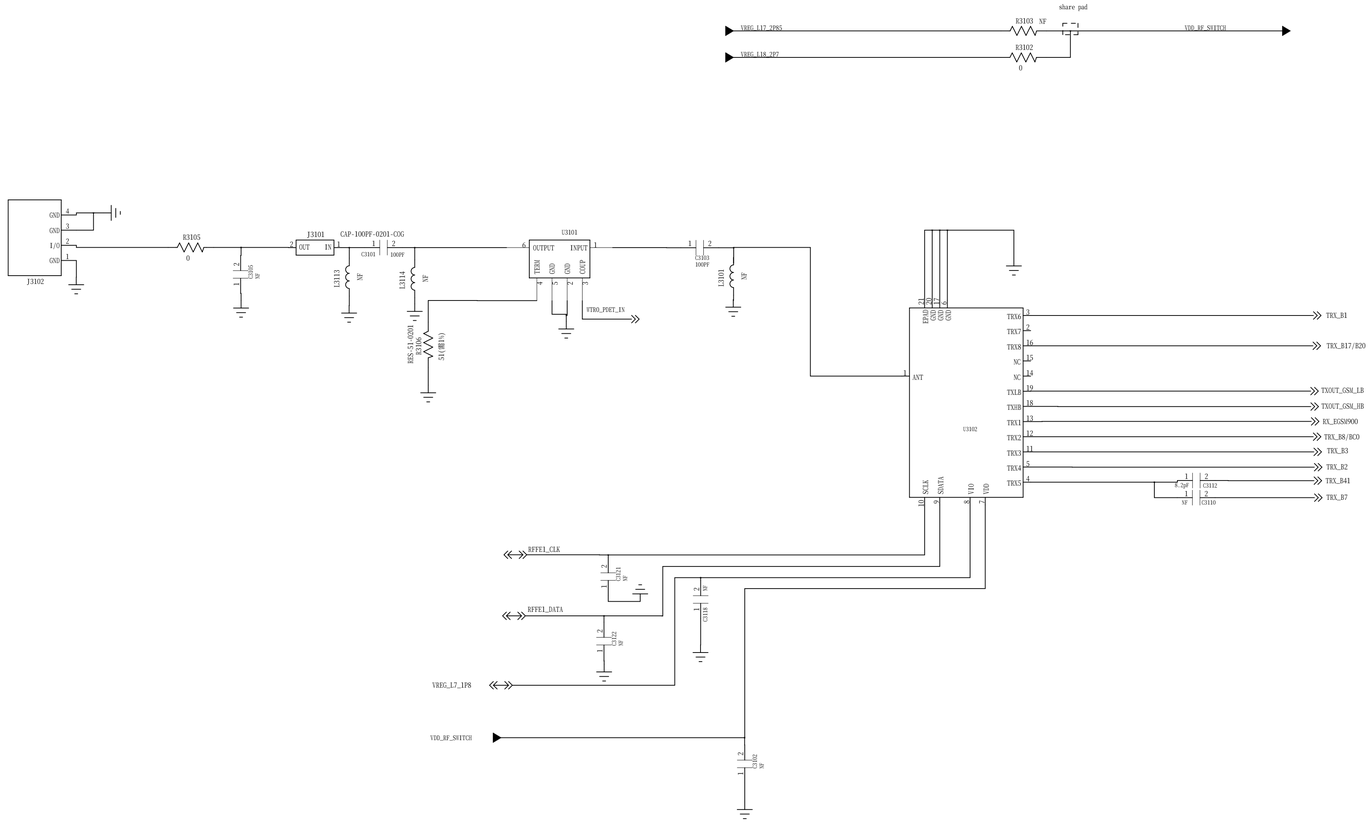

	ANT_SWITCH_CTL0	ANT_SWITCH_CTL1
TRX_B7		
TRX_B40		
QFE2320_ANT		
TRX_B38_41		

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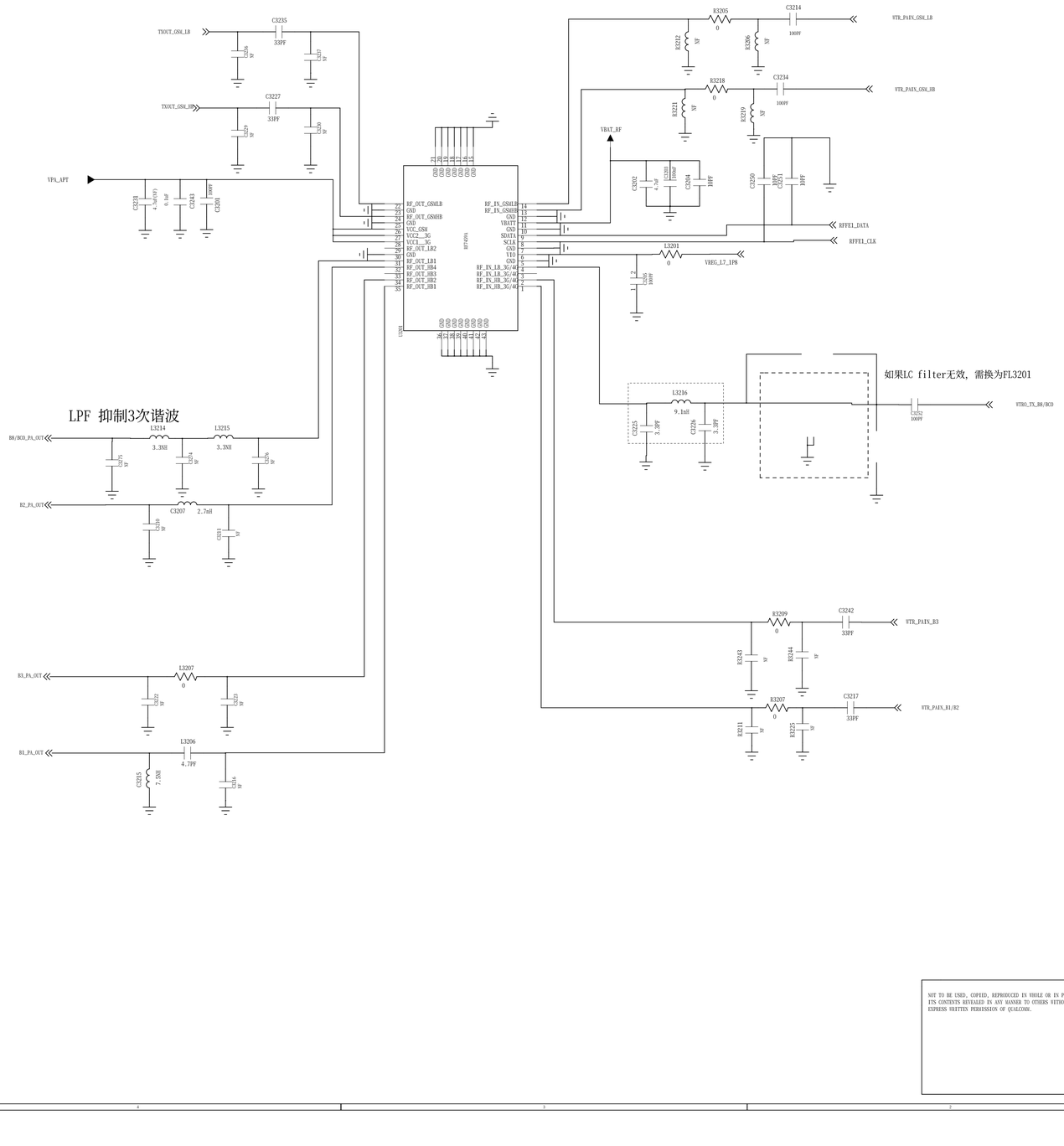
MMMB PA : B1,2,3,5,17 2G HB, 2G LB



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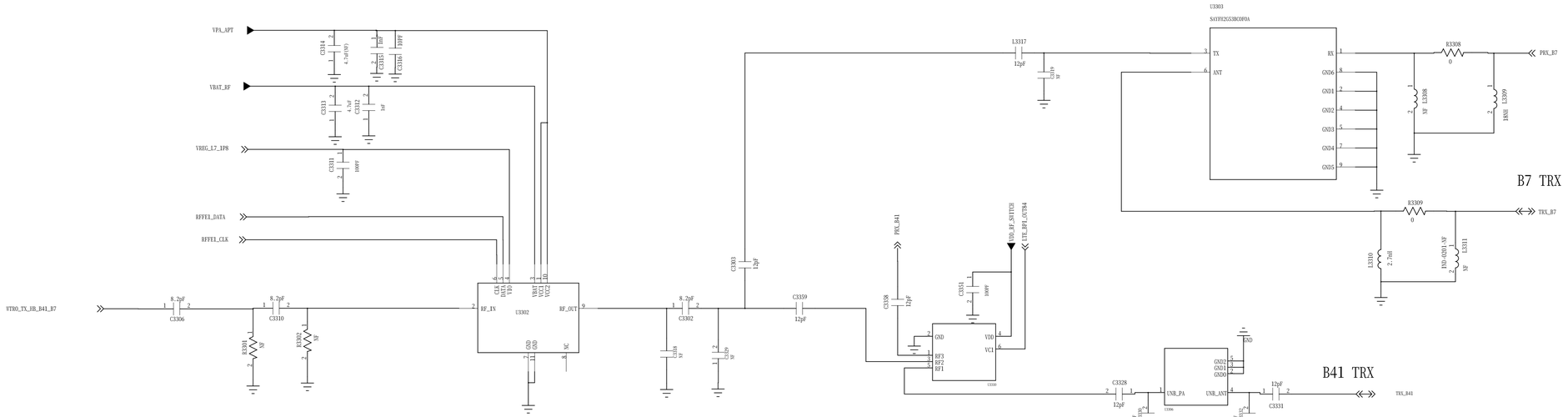

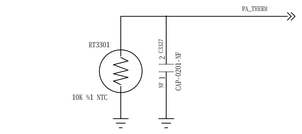

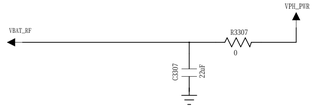

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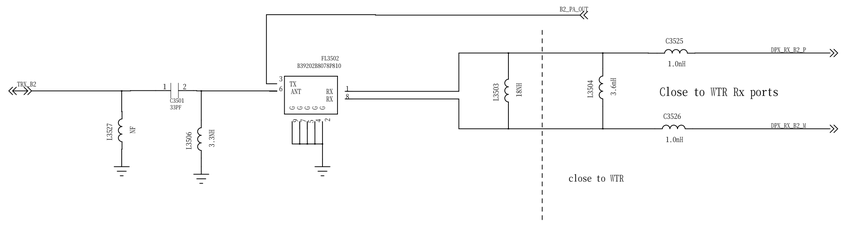

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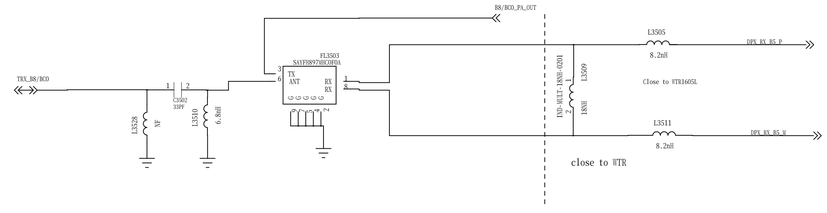

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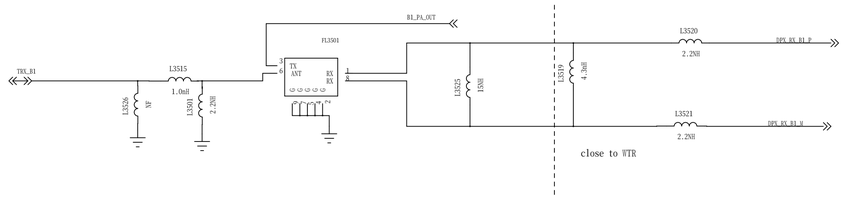

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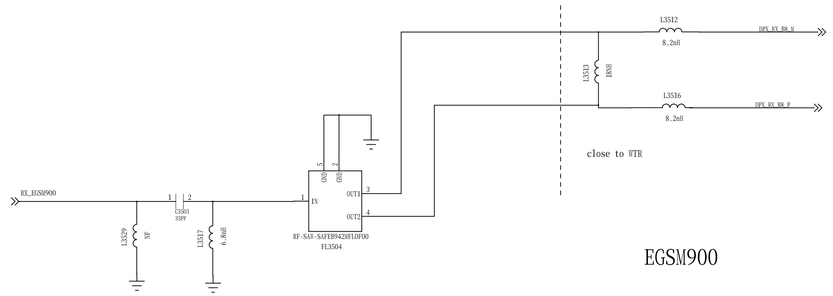

EGSM900

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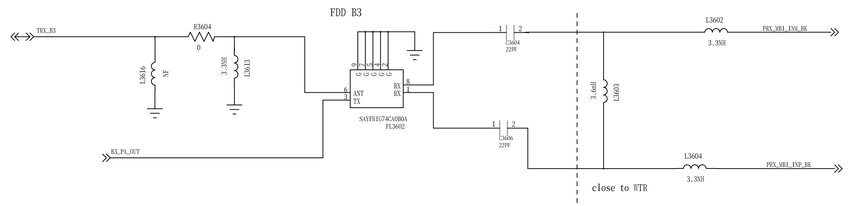

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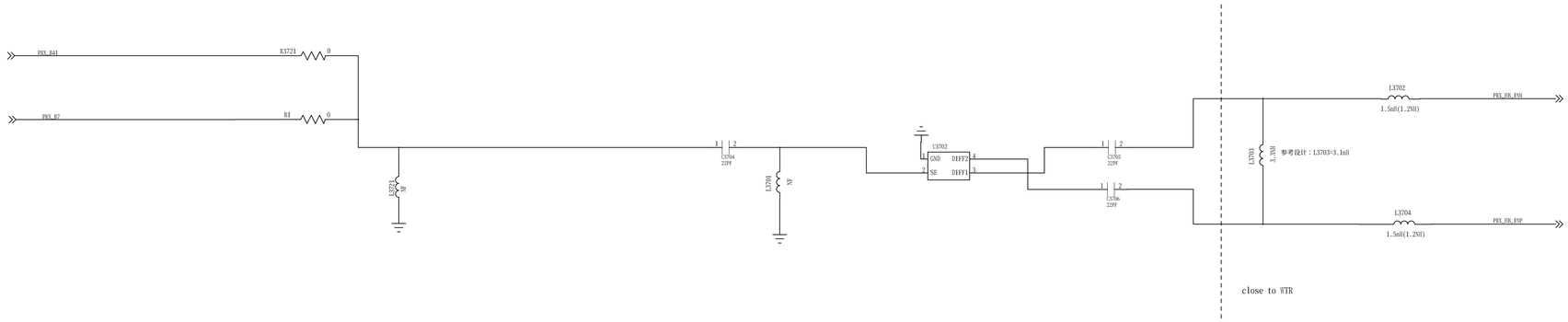

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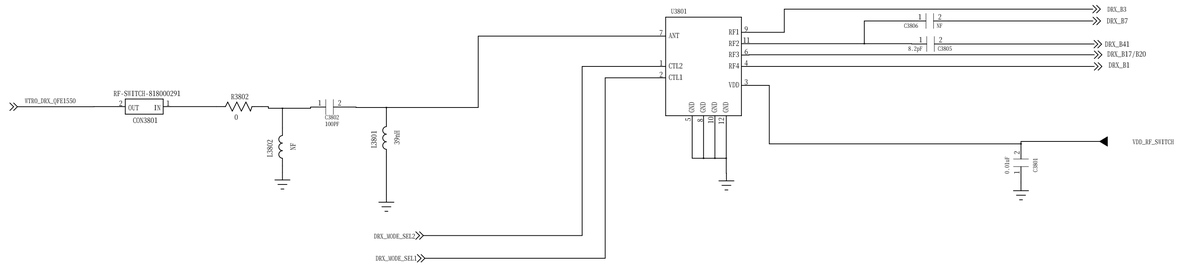

RX BAND SELECT LOGIC

	PORT	DRX_MODE_SEL0	DRX_MODE_SEL1	DRX_MODE_SEL2
DRX_B7	RF1			
DRX_B40	RF2			
DRX_B41	RF3			
DRX_B39_B1	RF4			
DRX_B3	RF6			

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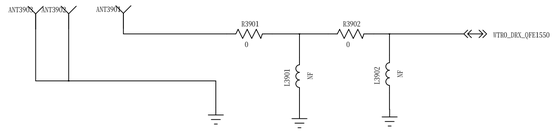

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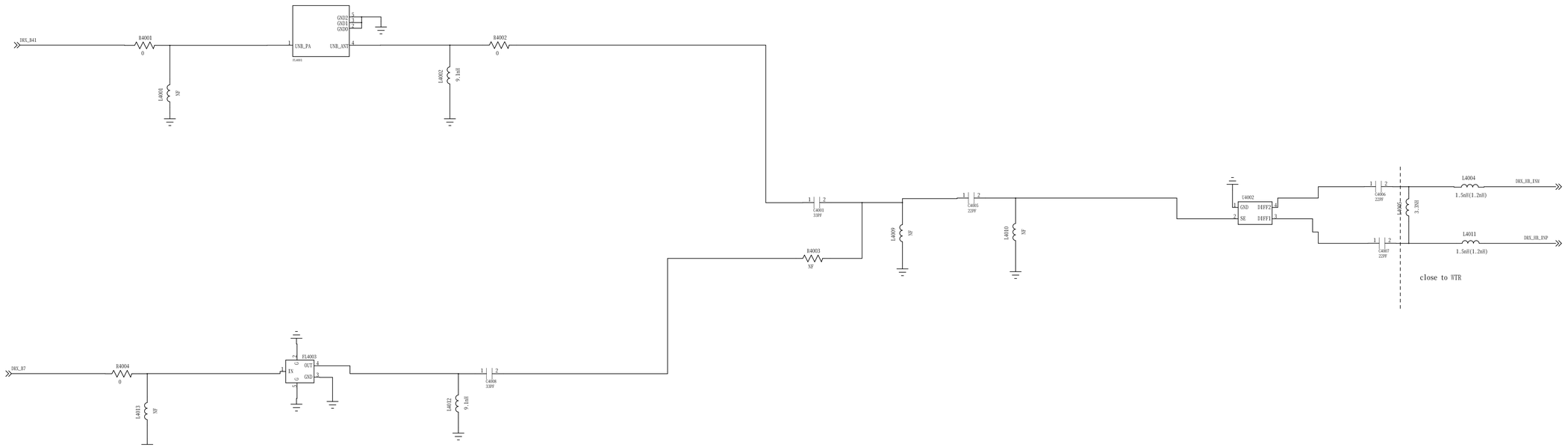

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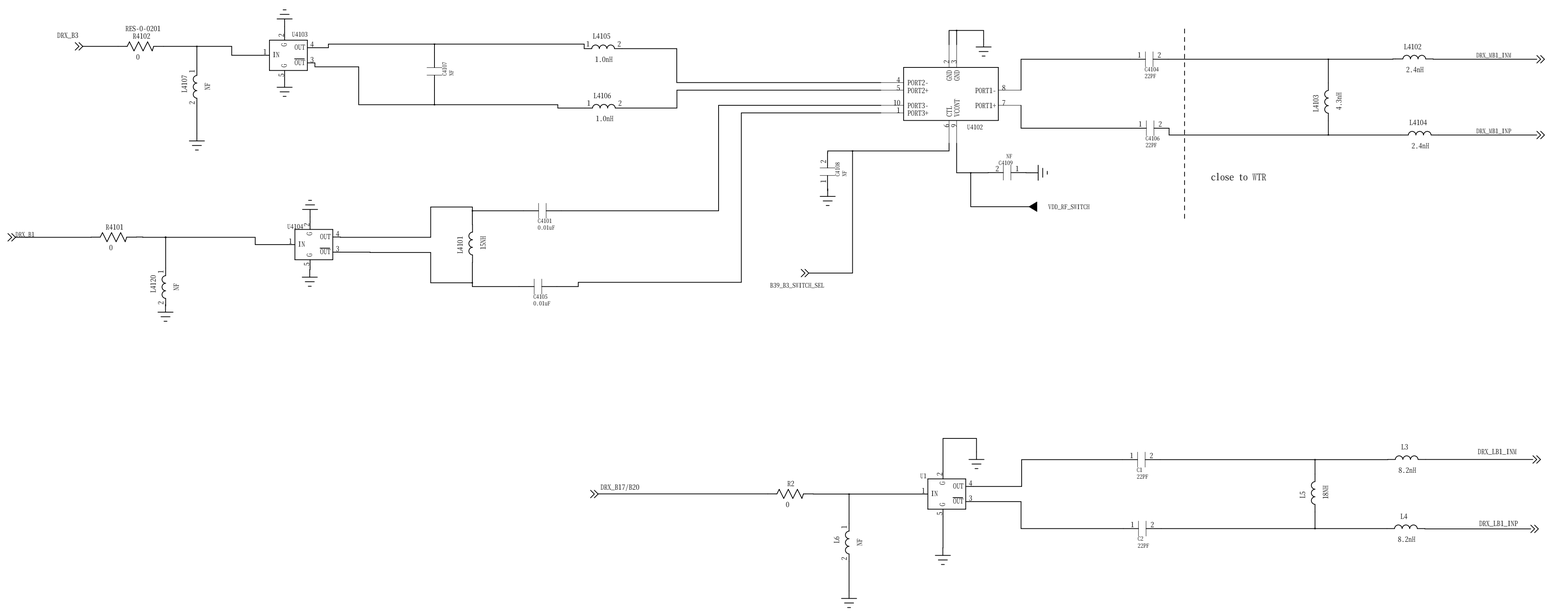

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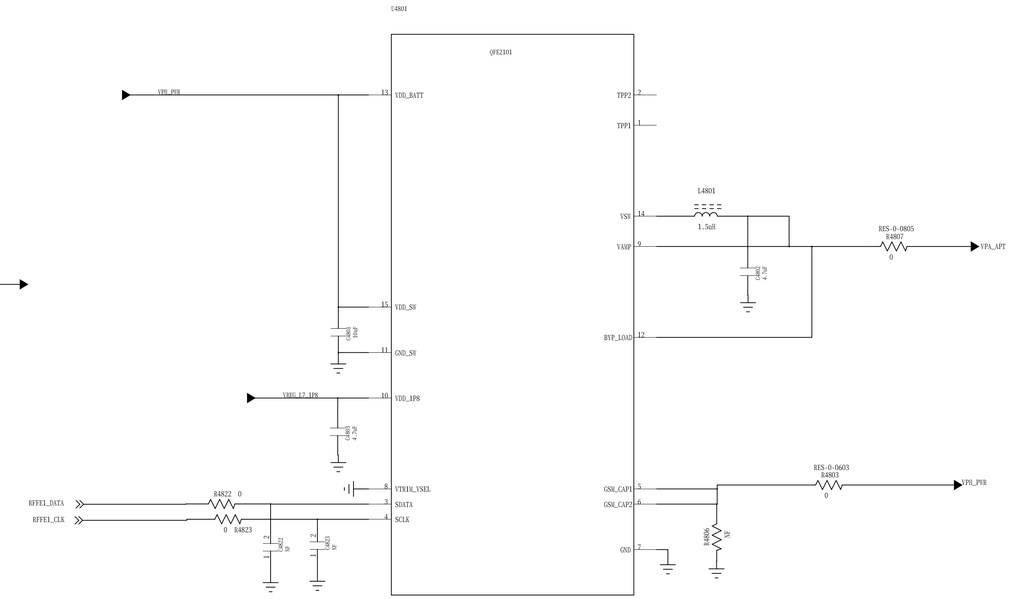
Note: Use a signle low impedance power plane/fill VPA for all ET PA VCC1/VCC2.

Refer to 80-NA681-91 rev.B or later revisions for more layout details.



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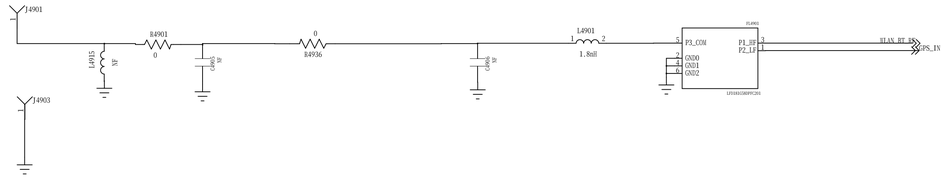

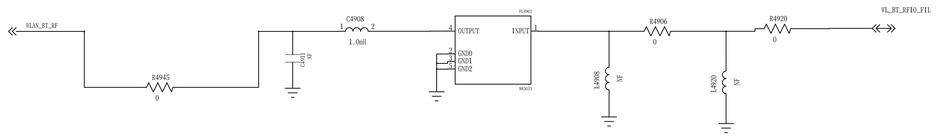

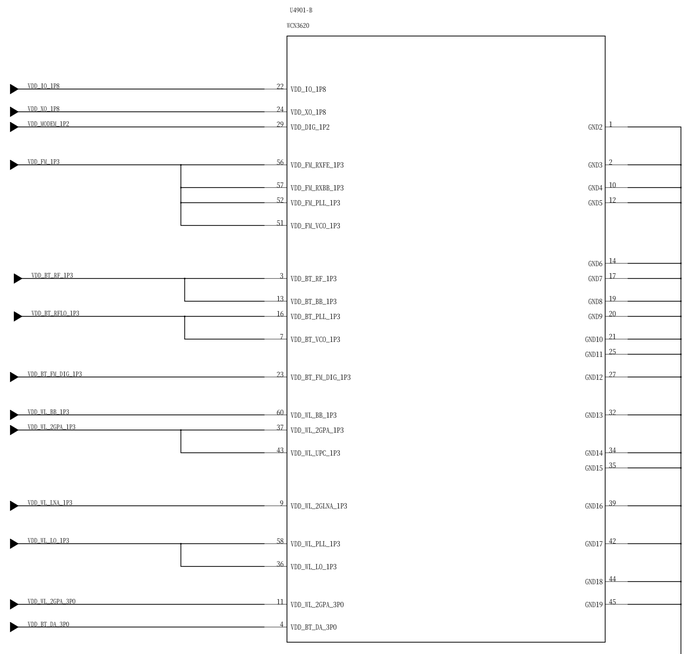

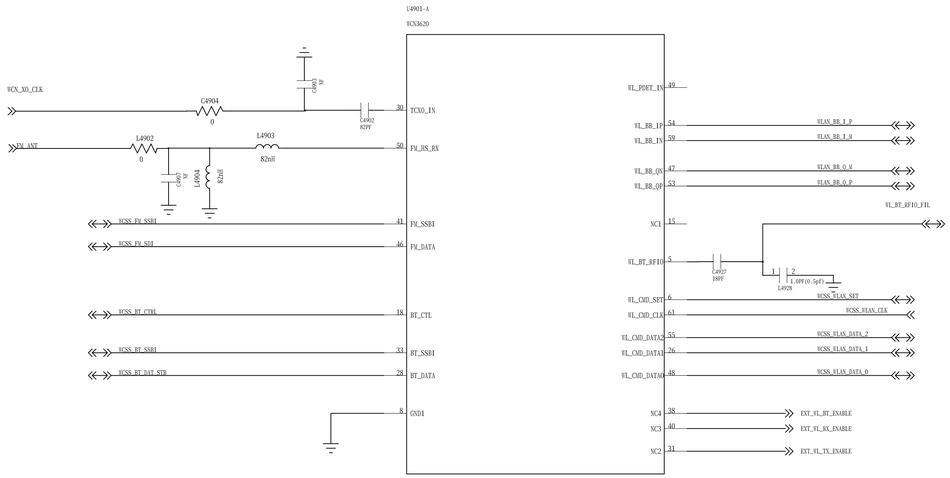

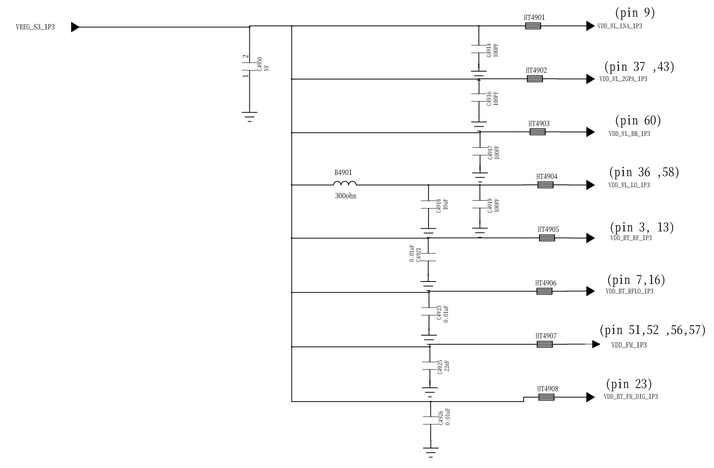

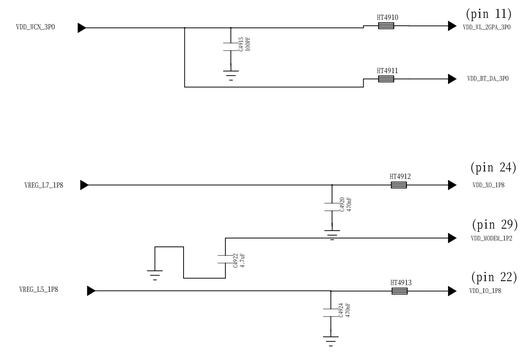

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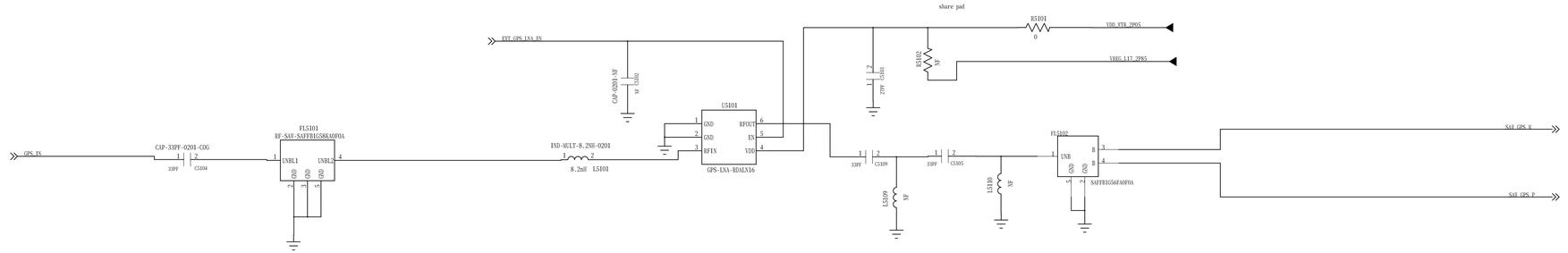

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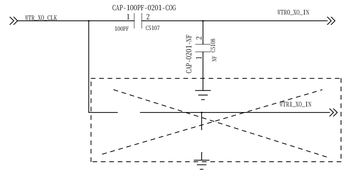

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